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Statistical Causal Inference

A Gentle Introduction to Advanced Analysis of Quantitative Data



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<https://julianfrattini.github.io/>



JulianFrattini/**bda4sci**

Public

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Julian Frattini // about me

2013 - 2019

Bachelor & Master of Science

 Technical University of Munich,
Munich, Germany

2020 – 2025

Ph.D. Student

 Blekinge Institute of Technology,
Karlskrona, Sweden

2025 - 2029

Postdoctoral Researcher

 Chalmers University of Technology,
Gothenburg, Sweden

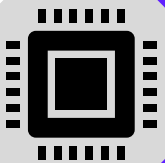


Julian Frattini // about my research



Good-enough Requirements Engineering

Quantifying the effect of quality factors of requirements artifacts on downstream software development activities



Hybrid-AI Software Engineering

Understanding the interface between traditional and data-driven software components in a hybrid software system.



Software Engineering Research Methodology

Improving the methods used in research to produce more rigorous and valid conclusions



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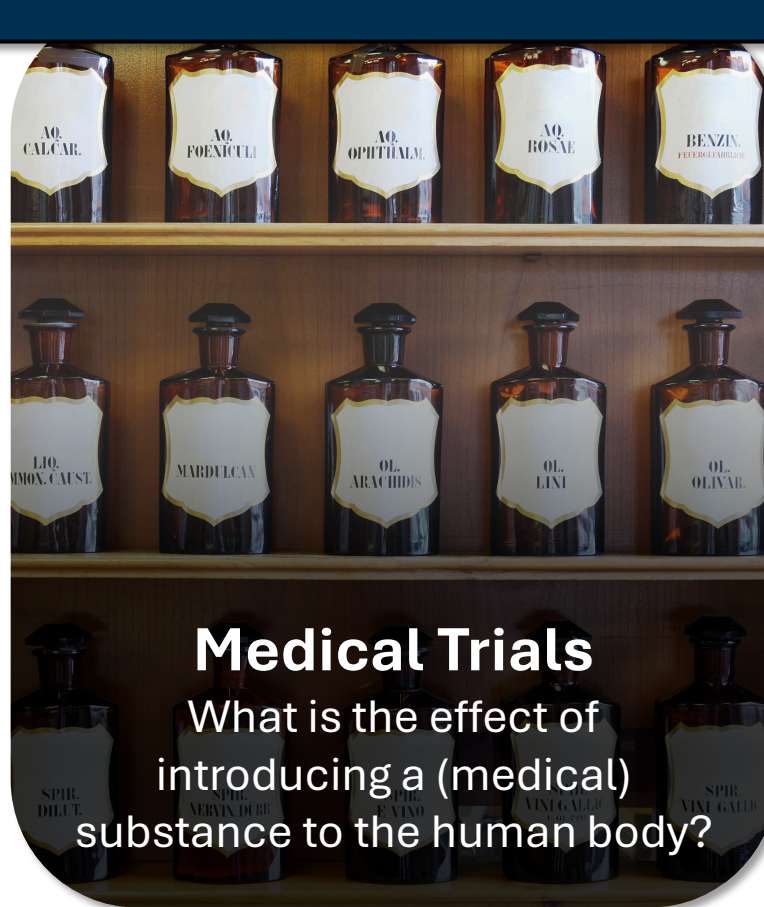
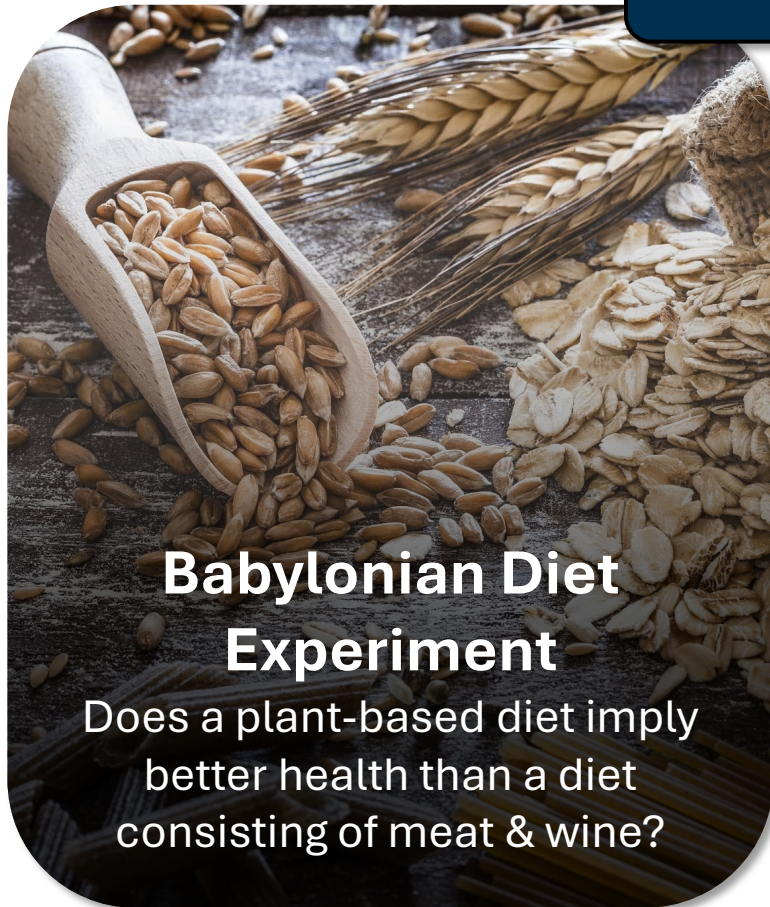
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Motivation

The Point of Science

The Point of Science

” Science informs decision-making policies.



Types of Scientific Studies



We can safely and ethically **control** the factor of interest (diet), i.e., assign one of two diets (plant- vs. meat-based) to all subjects of our study and record their health outcome.

→ (controlled) experimental studies



We cannot force subjects to smoke, we can only **observe** their natural behaviour (whether they already smoke or not) as well as their health outcome.

→ observational studies



But what if an observation that smokers have worse health is caused by another factor (e.g., social status or general health)?

Types of Scientific Evidence



Experimental Studies

If we only vary one particular factor (e.g., the diet), we can assume that any change in health **is caused** by this factor.

Changing that factor (i.e., switching ones diet) *should* show the same effect in any person.



Observational Studies

If several factors vary (e.g., social status, general health) in addition to the one of interest (i.e., smoking), we only know that any change in health **is correlated** with the main factor.

Changing that factor (i.e., stopping to smoke) *might not show* the same effect in any person as the other factors that remain the same may be responsible for the effect (i.e., change in health).

Correlation versus Causality



Correlation:

when we *observe* X,
we also observe Y



Causality:

when we *do* X,
we observe Y.

If we want to make a **positive impact** on the target audience of our research (i.e., software engineers), we need to provide **reliable decision-making policies** – which we can only derive from **causal** relationships, not correlations.



We cannot infer causality from data alone – we must also know the process that created that data.¹

¹we'll revisit this statement later

Limitations of Experiments

There are several reasons why we cannot investigate any phenomenon with a controlled experiment, as they are



Expensive

Recruiting eligible participants is difficult



Unrealistic

Experiments perturb the natural context



Impossible

Some factors are impossible to control



Assumed

The validity of experimental results depends on several conditions



Still, we can draw causal conclusions from non-experimental studies using the principles of **statistical causal inference**.

Goals of this Lecture

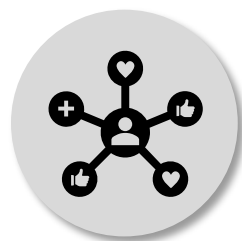


Convincing you of the **value of causal conclusions**



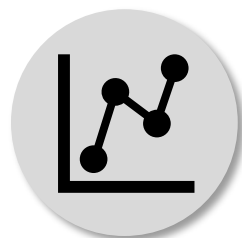
Learning the fundamentals of **drawing causal conclusions** from quantitative data collected in **observational studies**

Learning Outcomes



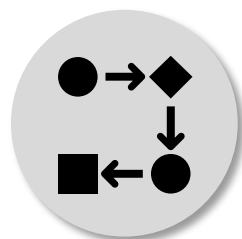
Causal Modelling

Communicating transparent causal assumptions in graphical models



Statistical Causal Inference (SCI)

Drawing reliable conclusions from quantitative, observational data



Causal Workflow

Structured process for performing statistical causal inference in an empirical study



Causal Modelling

Concisely Framing Causal Assumptions

Basic Terminology

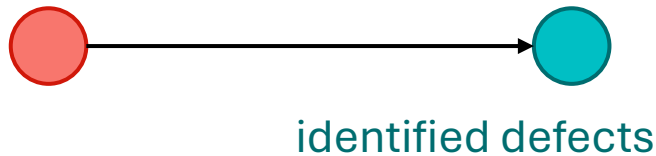
“We study the effect of different **document reading techniques** on the **number of identified defects** in **Bachelor- and Master-level** students.”

- **Factors**: variables that associate measurements with a construct (e.g., document reading technique, identified defects, student level)
- **Treatment/Exposure**: main independent factor of interest
- **Outcome/Response**: main dependent factor(s) of interest
- **Phenomenon**: main causal effect of interest (**treatment** → **outcome**)

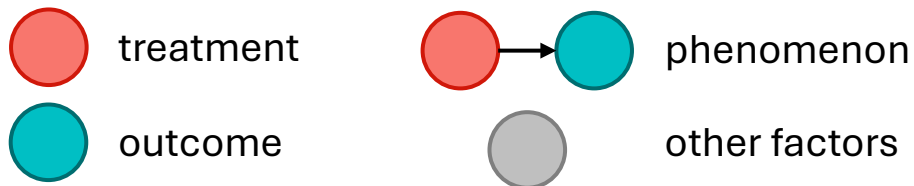
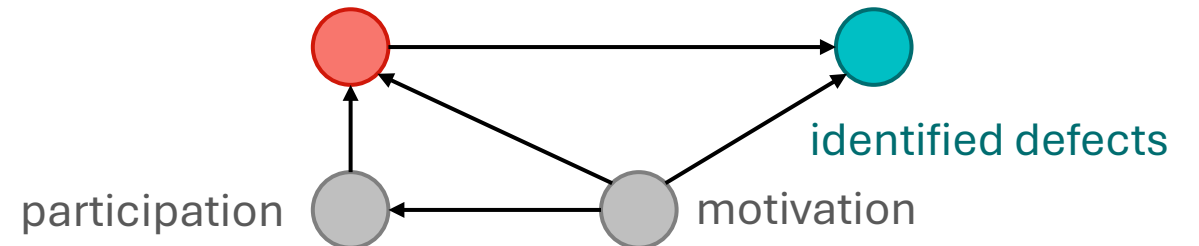
Causal Modeling

Directed, acyclic graphs (DAGs) can represent causal assumptions, where *nodes represent variables* and *edges represent assumed/potential causal relationships* between those variables.

document reading technique



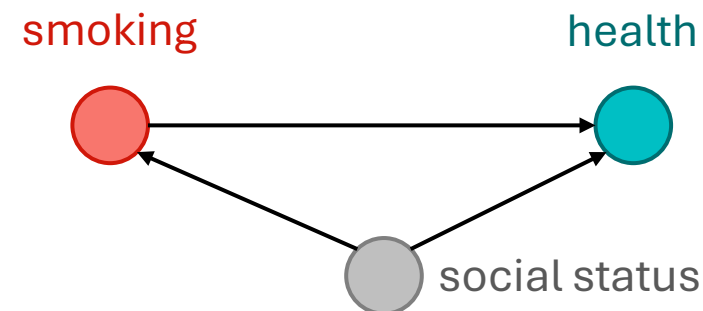
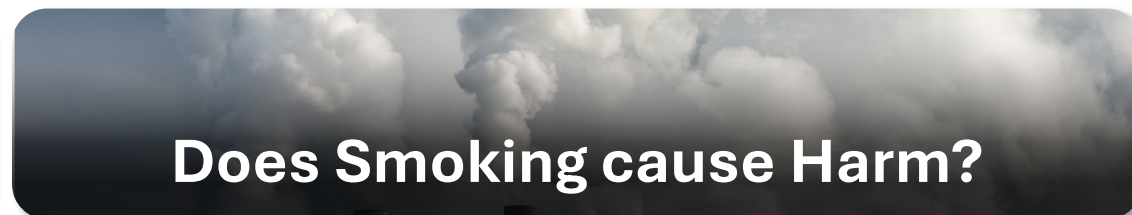
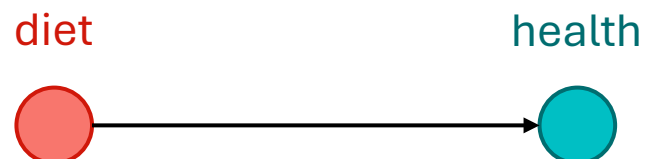
document reading technique



DAGs are subjective and represent the view of a domain expert.

Causal Modeling

DAGs allow visualizing causal hypotheses clearly:



” In experimental studies, no factor influences the **exposure** (other than “randomness”).

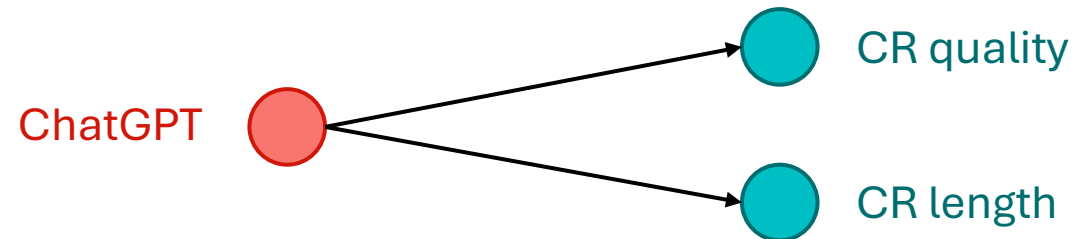
Exercise: Formulating a DAG

Derive a causal DAG from the following description:

“Researchers are interested in knowing if using ChatGPT affects the quality and length of code reviews at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews.”

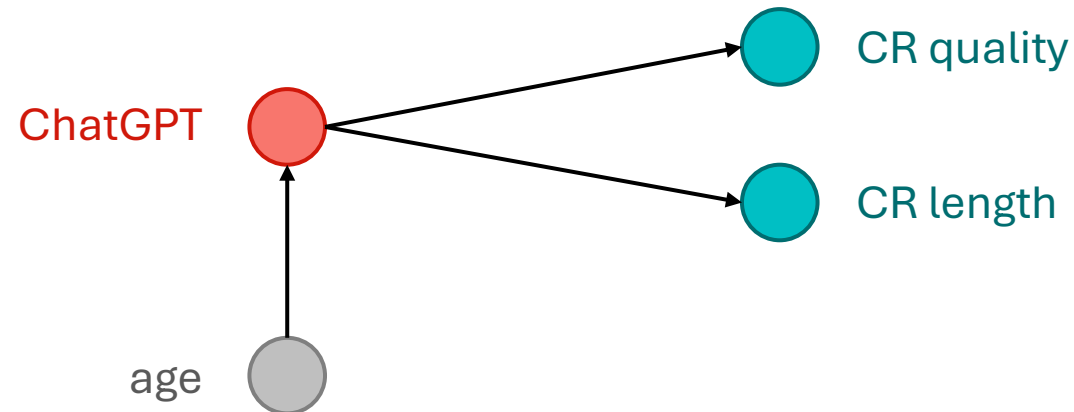
Solution: Formulating a DAG

“Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews.”



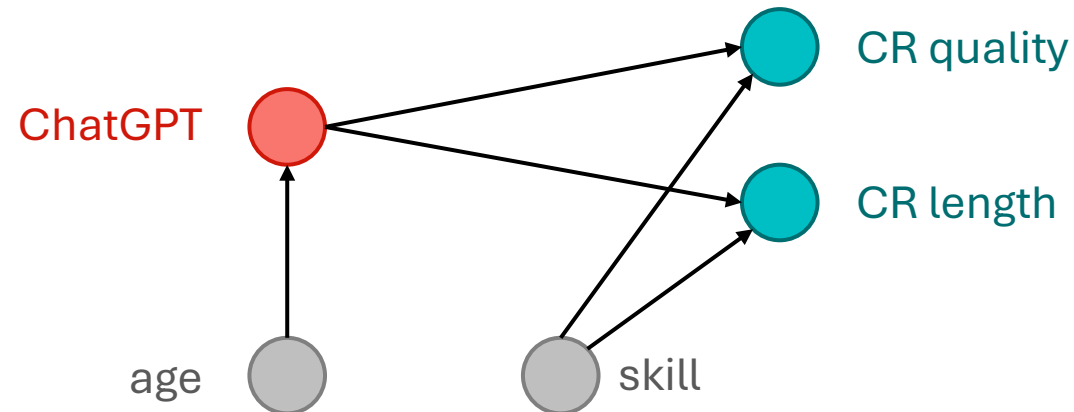
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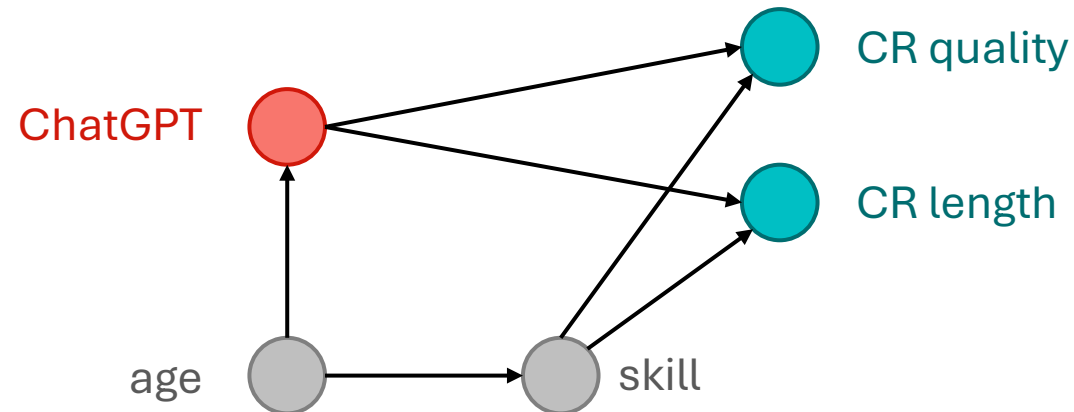
Solution: Formulating a DAG

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Solution: Formulating a DAG

“Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that **younger developers** use ChatGPT more often. On the other hand, they know that **more skilled developers** generally produce more concise (i.e., shorter) and better code reviews.”



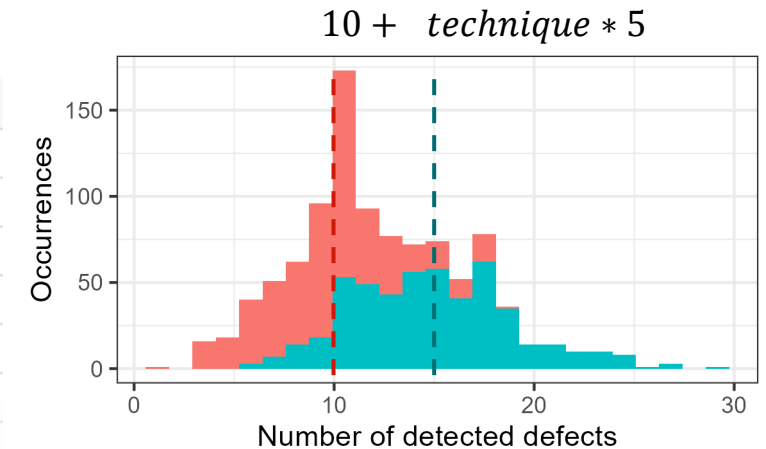
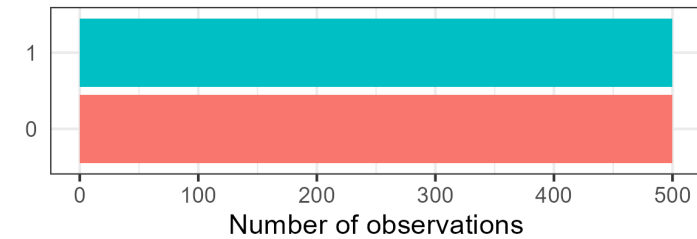
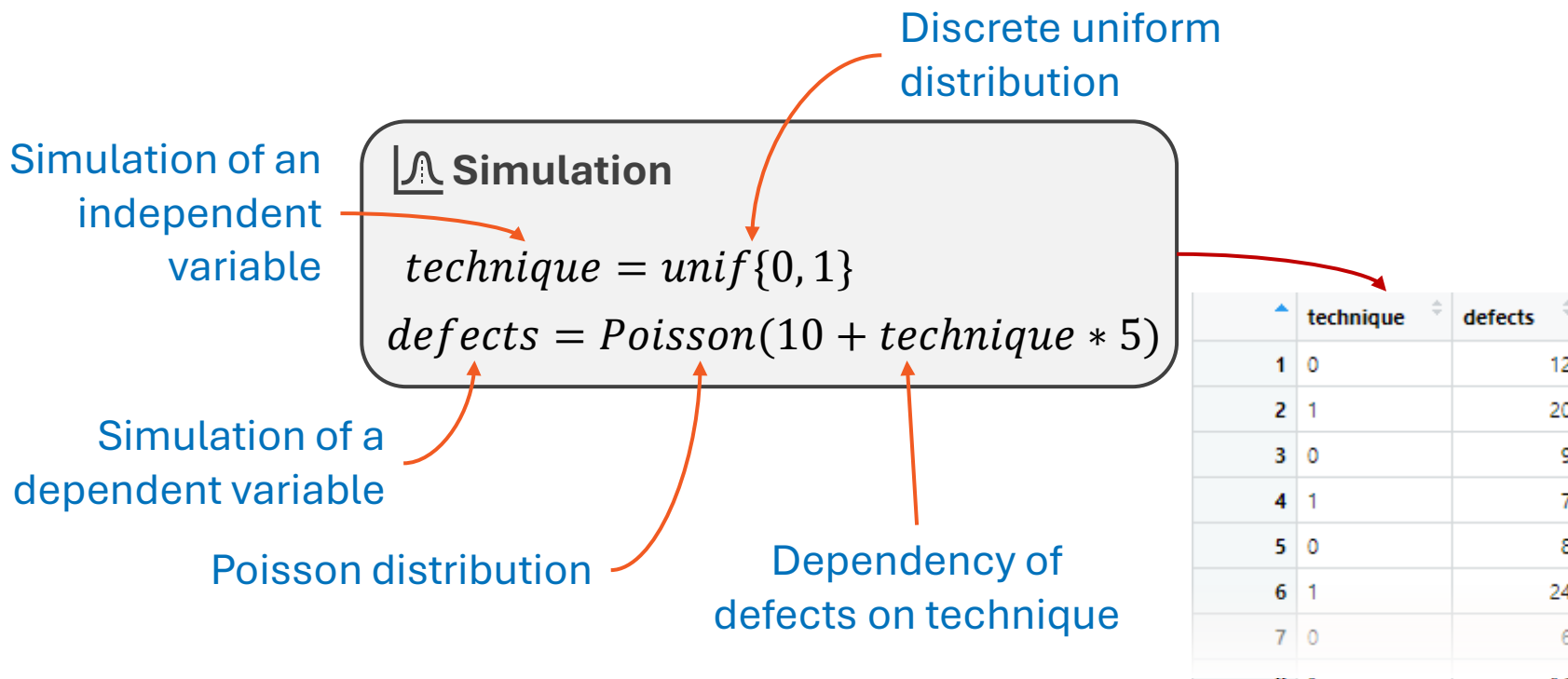


Fundamentals & Notation

Syntax and Pedagogics of this Lecture

Simulations

To demonstrate the process of SCI, we **simulate** data sets. This gives us control over phenomena, i.e., a reliable "ground truth."



Technique ■ Baseline (0) ■ Treatment (1)

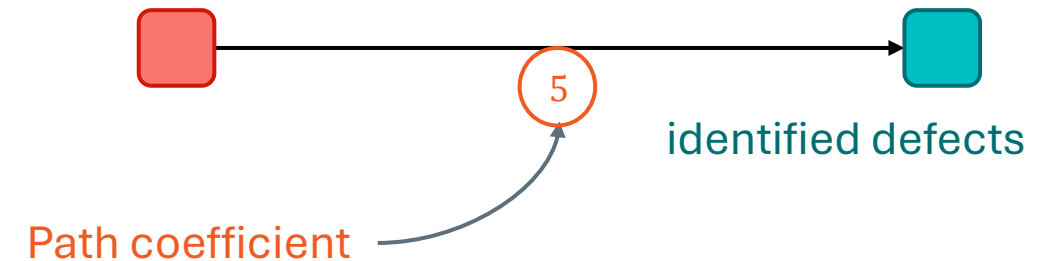
Causal Modeling and Simulations

DAGs are typically used to formalize *assumed* causal relationships. They can also be used to represent **simulated** causal relationships.

Simulation

```
technique = unif{0, 1}  
defects = Poisson(10 + technique * 5)
```

document reading technique



”

Changing the **cause** by one unit causes the **effect** to change by <path coefficient> units. It is, therefore, our variable of interest representing effect strength.

Data Analysis Tool: (Linear) Modeling

+ more accessible

- less granular

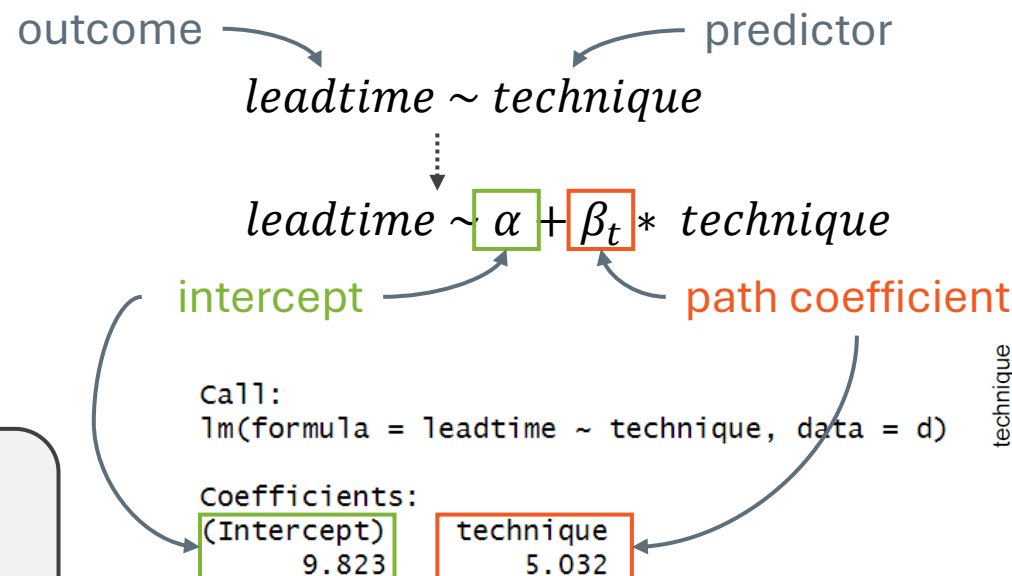
more difficult to use and apply -

more granular +

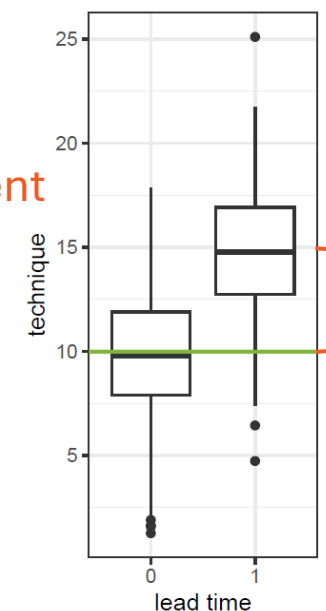
Null-Hypothesis Significance Test

e.g., Student's t-test,
Mann-Whitney U Test

(Generalized) Linear (Mixed) Model



Bayesian Data Analysis

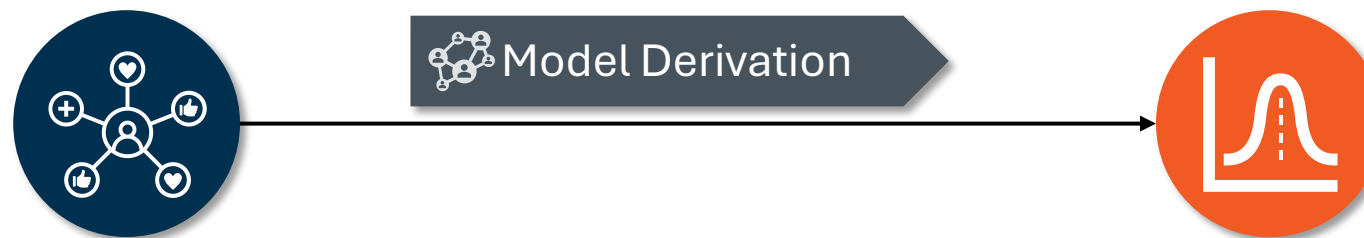


Simulation

$technique = unif\{0, 1\}$

$leadtime = N(10 + technique * 5, 3)$

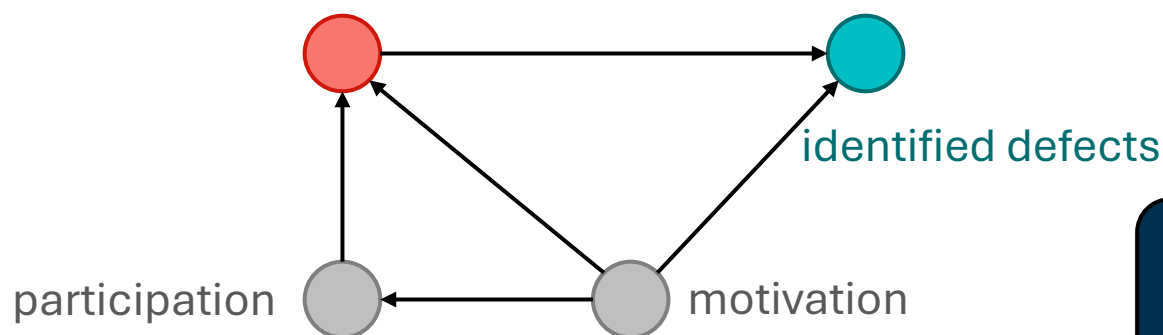
From causal to statistical Models



Causal Models
encode causal
assumptions

Statistical Models
implement inference

document reading technique



-----> $defects \sim technique + motivation$

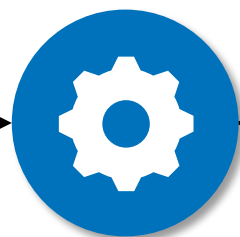
” How to derive statistical from causal models is core to statistical causal inference.

Pedagogy of this Tutorial



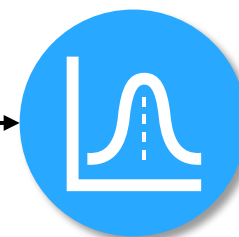
1. Ground Truth

Simulating data based on defined causal relationships



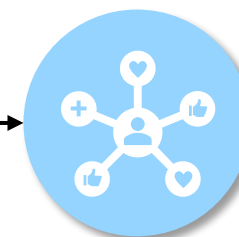
2. Regression

Deriving statistical models and running regression analyses



3. Comparison

Comparing the results of the analysis with the ground truth



4. Inference

Deducing obtainable knowledge from the analysis results

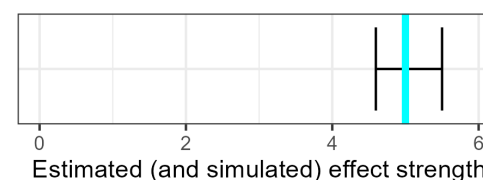
document reading
technique



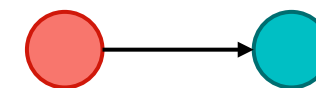
identified defects



$m: \text{defects} \sim 1 + \text{technique}$



document reading
technique



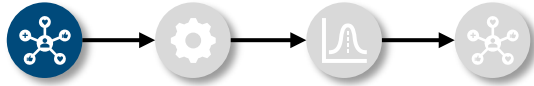
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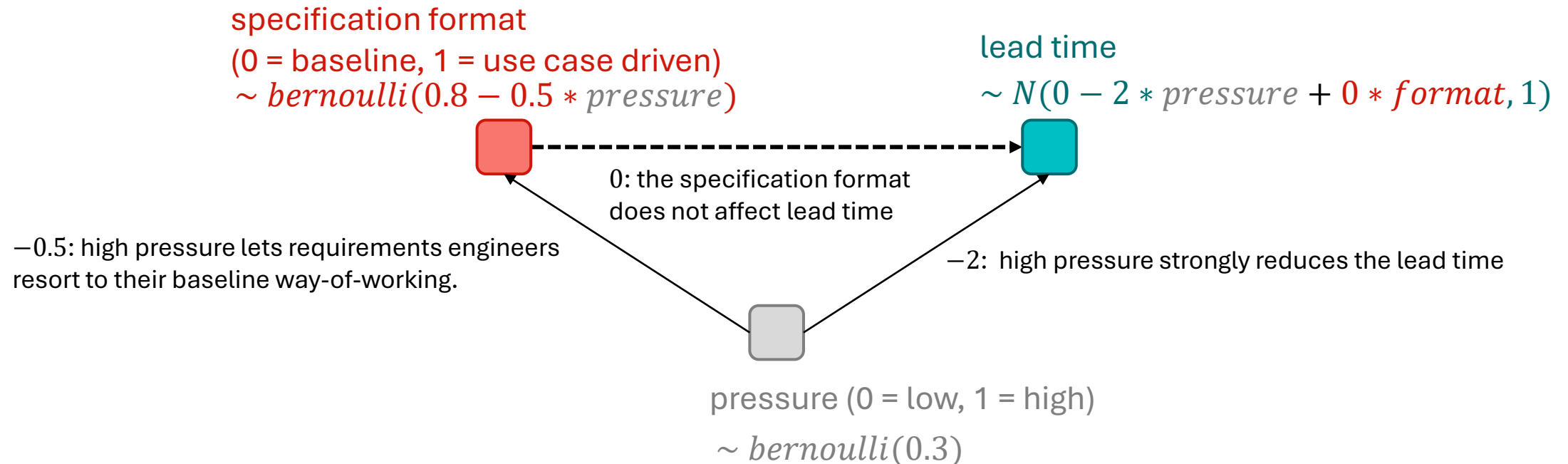
Statistical Causal Inference

Confounders and Mediators

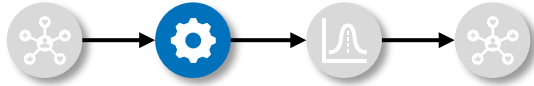
Confounding: Common Causes



“The company pushed for a use case driven requirements specification approach. We investigate whether the **specification format** of a requirement decreased the normalized **lead time**.”

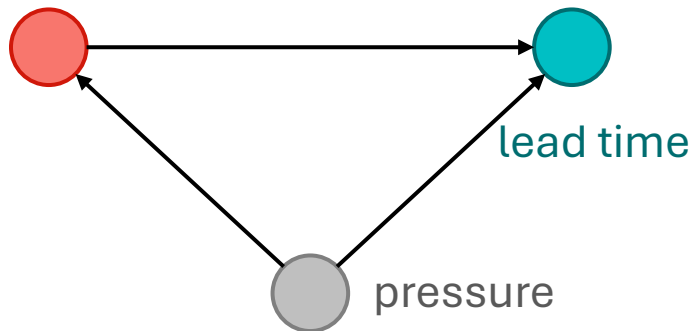


Confounding: Common Causes



“The company pushed for a use case driven requirements specification approach. We investigate whether the **specification format** of a requirement decreased the normalized **lead time**.”

specification format

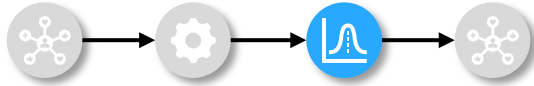


Model Derivation

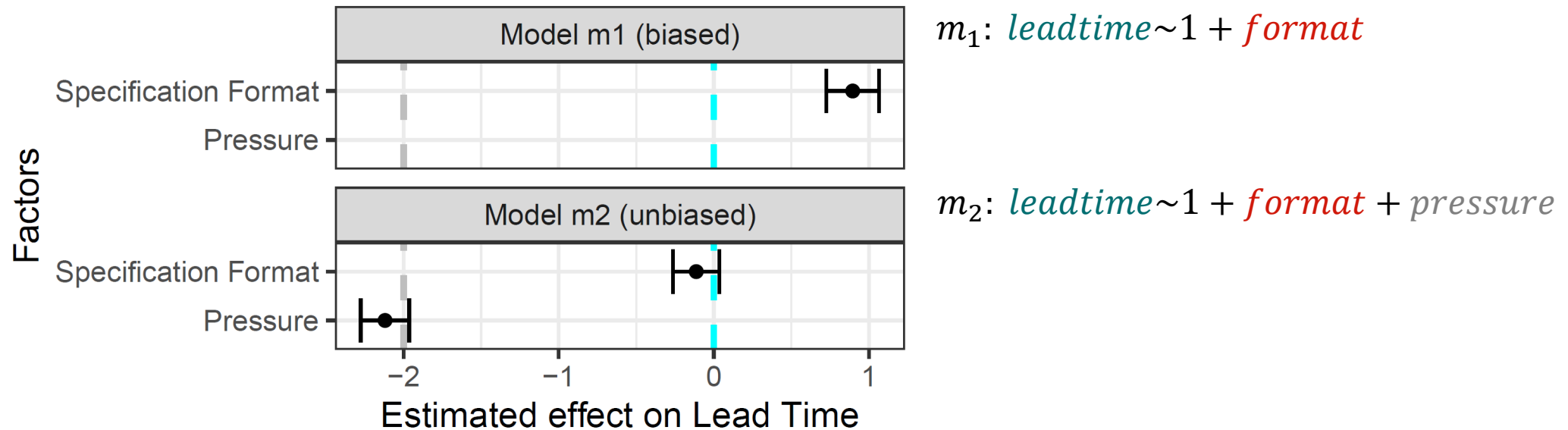
$$m_1: \text{leadtime} \sim 1 + \text{format}$$

$$m_2: \text{leadtime} \sim 1 + \text{format} + \text{pressure}$$

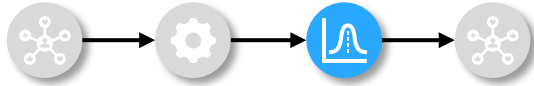
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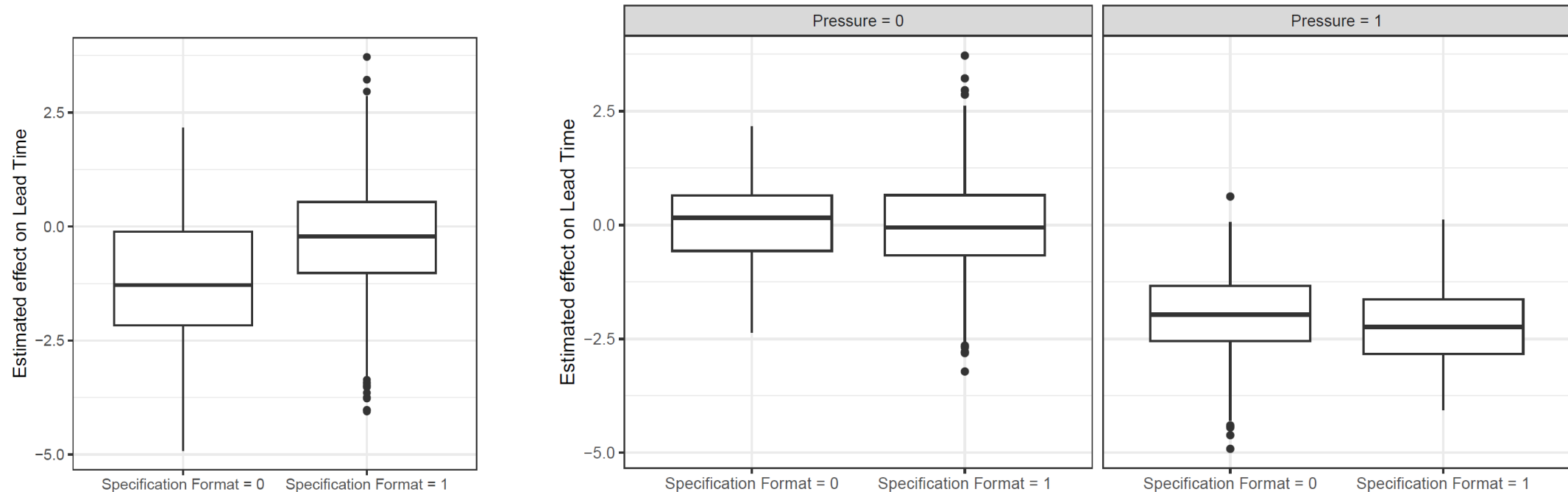
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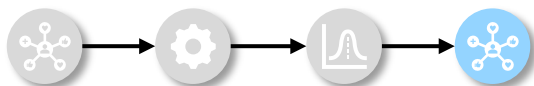
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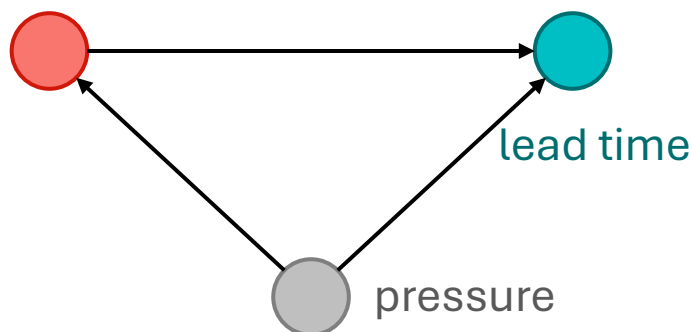


Confounding: Common Causes



“The company pushed for a use case driven requirements specification approach. We investigate whether the **specification format** of a requirement decreased the normalized **lead time**.”

specification format



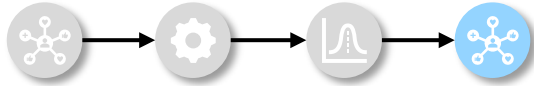
 Model Derivation

$$m_1: \text{leadtime} \sim 1 + \text{format}$$

$$m_2: \text{leadtime} \sim 1 + \text{format} + \text{pressure}$$

” If both **exposure** and **outcome** have a common cause, we need to statistically adjust for (i.e., control) the confounder to de-bias the effect of interest.

Confounding: Common Causes

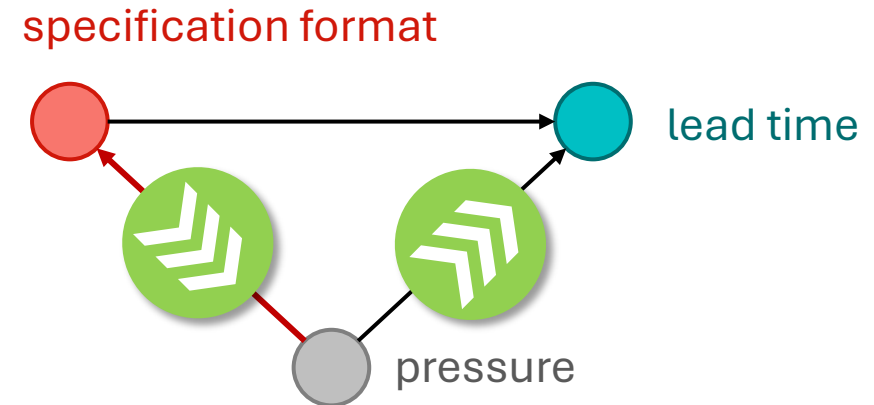


Whenever **specification format = 0**, the likelihood that **pressure = 1** is much higher which causes a **lower lead time**.

”

“Whenever **specification format = 0**, the likelihood that **pressure = 1** is much higher” is an association, but not causal in this direction!

Remember:  observing $\neq$  doing

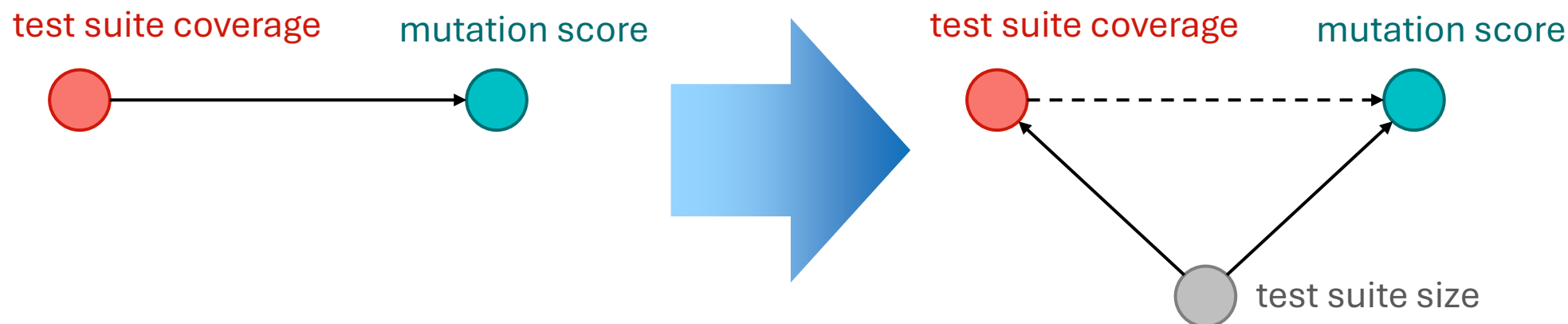


When *not* controlling the confounder, “information flows” from the cause through the confounder to the effect in a non-causal direction.

Practical Example of Confounding

“Coverage Is Not Strongly Correlated with Test Suite Effectiveness”

We found that there is a low to moderate correlation between coverage and effectiveness when the number of test cases in the suite is controlled for. In addition, we found that

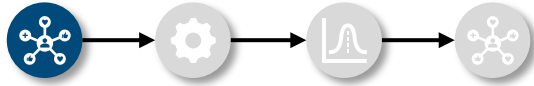


<http://dx.doi.org/10.1145/2568225.2568271>

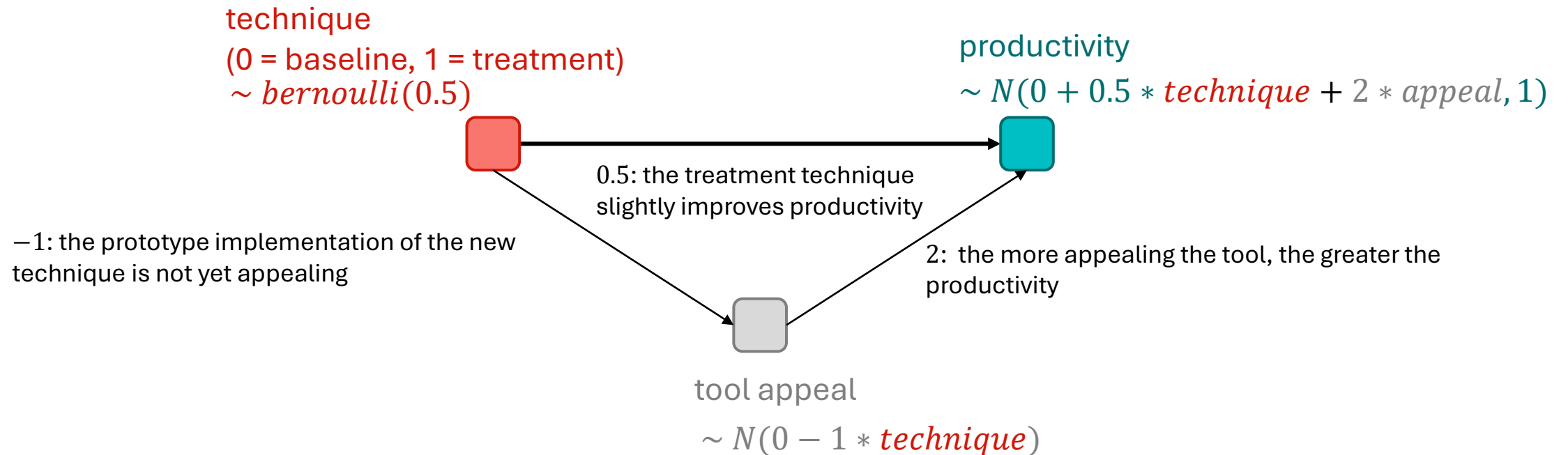
Inozemtseva, L., & Holmes, R. (2014, May). Coverage is not strongly correlated with test suite effectiveness. In *Proceedings of the 36th international conference on software engineering* (pp. 435-445).

Most Influential Paper at ICSE'24

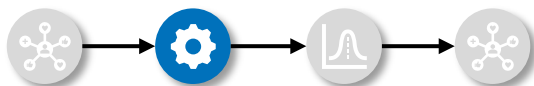
Mediation: Pipes



“The company tested a new **technique** in their requirements engineering phase that is enabled by a **prototype tool**. We investigated whether this technique improved the engineers’ **productivity**.”

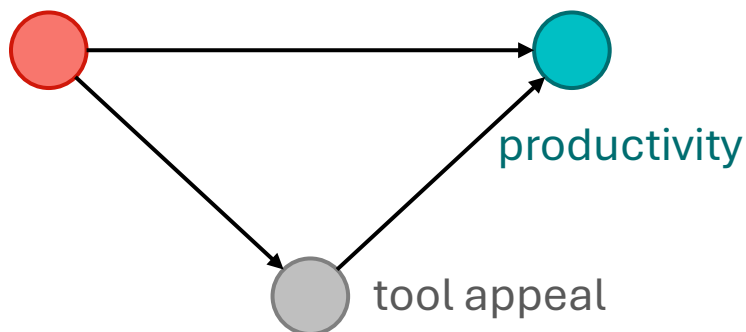


Mediation: Pipes



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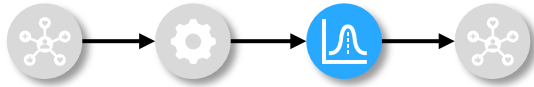
technique



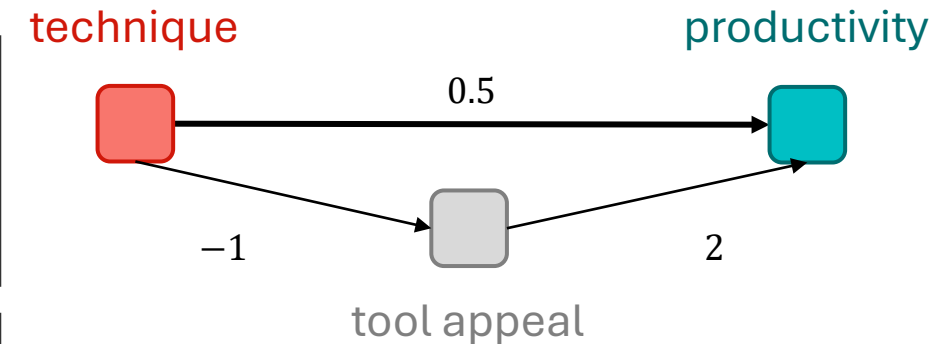
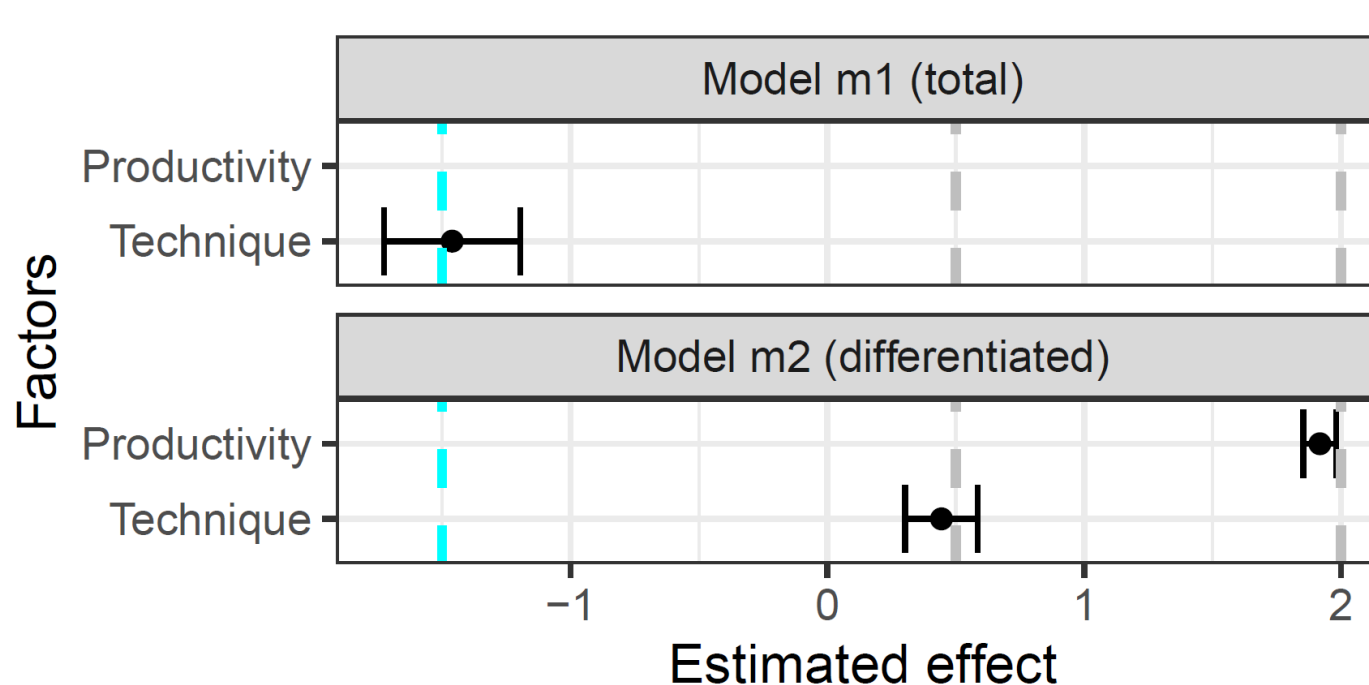
$$m_1: productivity \sim 1 + technique$$

$$m_2: productivity \sim 1 + technique + appeal$$

Mediation: Pipes



“The company tested a new **technique** in their requirements engineering phase that is enabled by a **prototype tool**. We investigated whether this technique improved the engineers’ **productivity**.”

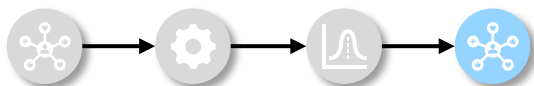


Direct effect: 0.5

Indirect effect: $-1 * 2 = -2$

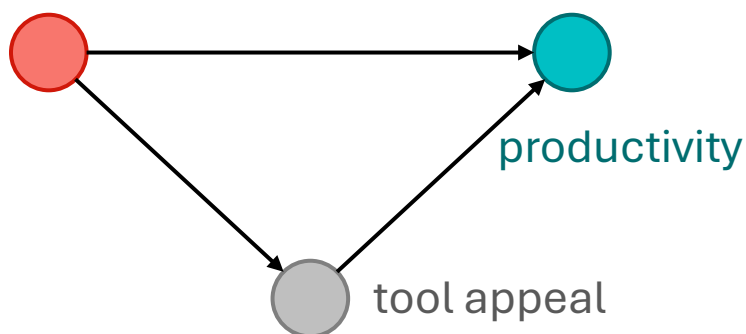
Total effect: $0.5 - 2 = -1.5$

Mediation: Pipes



Both statistical models are causally unbiased. The choice, which of them to prefer depends on whether one is interested in the direct or total effect of the treatment.

technique

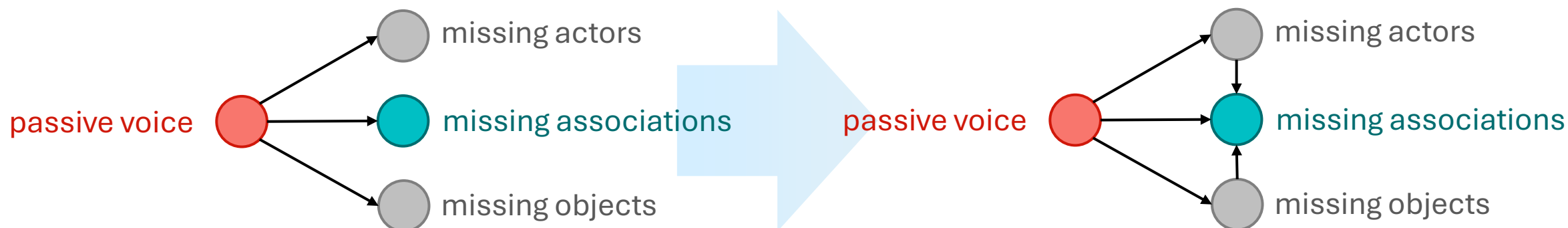


$m_1: productivity \sim 1 + technique$

$m_2: productivity \sim 1 + technique + appeal$

” If the effect of the **treatment** on the **mediator** is unstable, it is advisable to differentiate the direct from the indirect effect (i.e., $m_2 > m_1$).

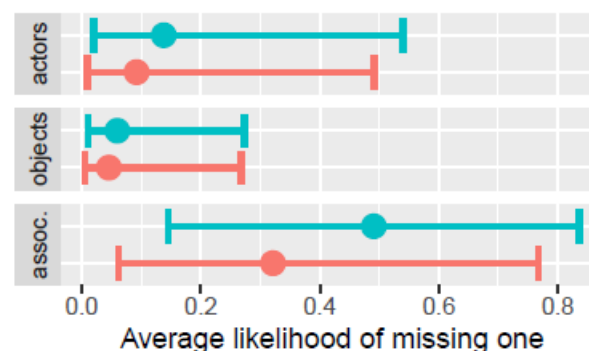
Practical Example of Mediation



Mann-Whitney test:

passive voice → **missing associations**

$$p = 0.03 < 0.05 = \alpha$$



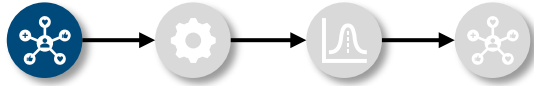
<https://doi.org/10.1145/2652524.2652554>

Femmer, H., Kučera, J., & Vetrò, A. (2014, September). On the impact of passive voice requirements on domain modelling. In *Proceedings of the 8th ACM/IEEE international symposium on empirical software engineering and measurement* (pp. 1-4).

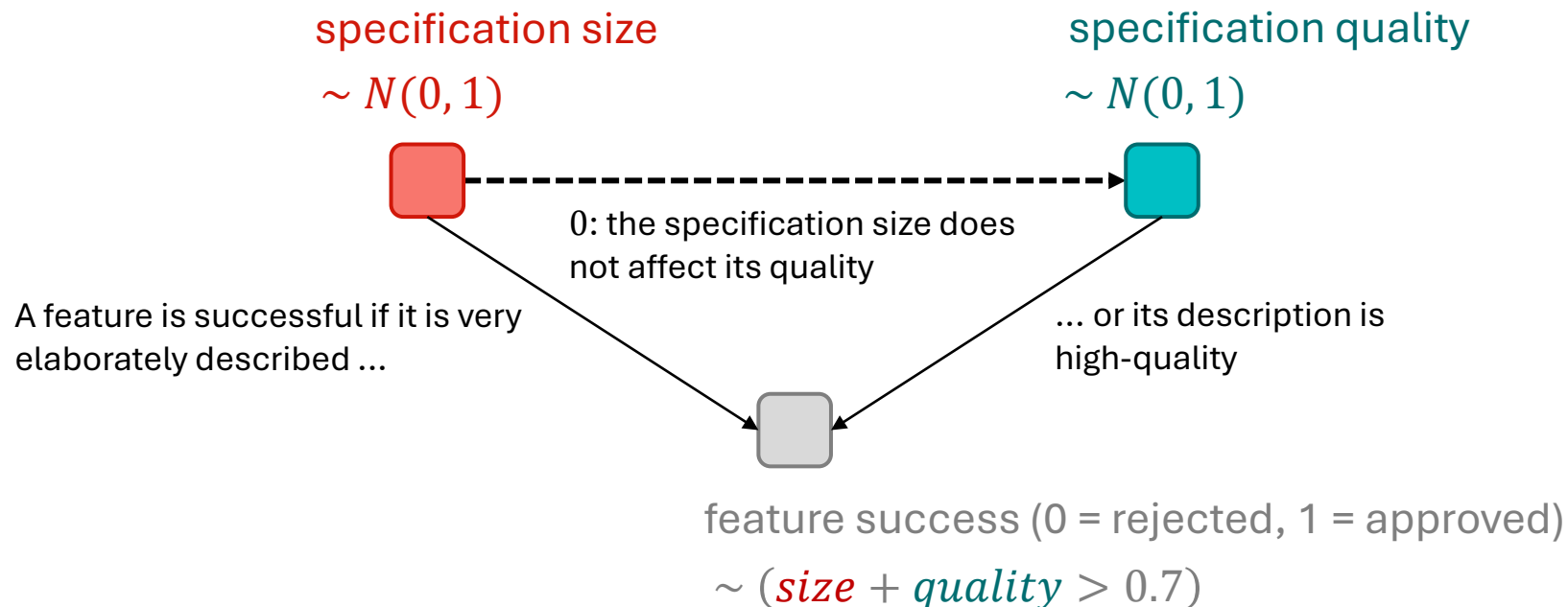
<https://doi.org/10.1145/3643664.3648211>

Frattini, J., Fucci, D., Torkar, R., & Mendez, D. (2024, April). A second look at the impact of passive voice requirements on domain modeling: Bayesian reanalysis of an experiment. In *Proceedings of the 1st IEEE/ACM International Workshop on Methodological Issues with Empirical Studies in Software Engineering* (pp. 27-33).

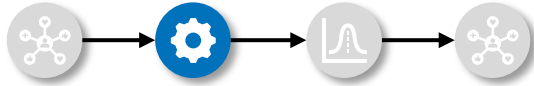
Colliding: Common Effects



“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”

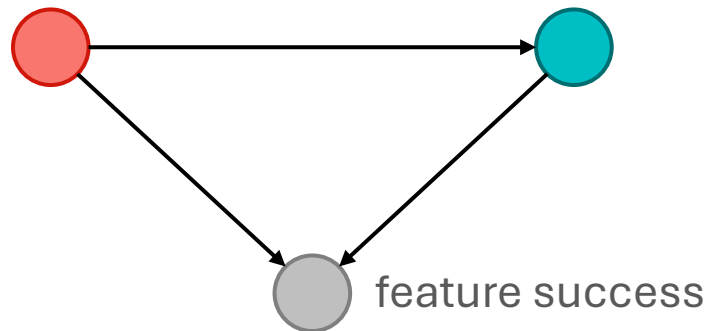


Colliding: Common Effects



“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”

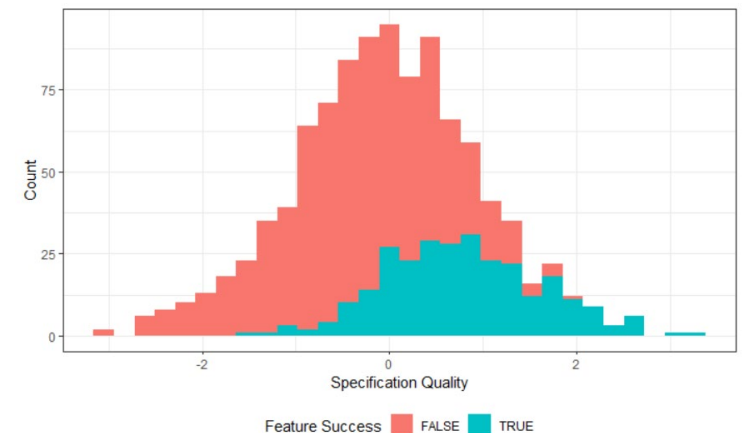
specification size specification quality



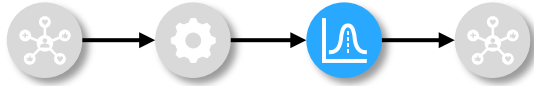
 Model Derivation

$m_1: \text{quality} \sim 1 + \text{size} + \text{success}$

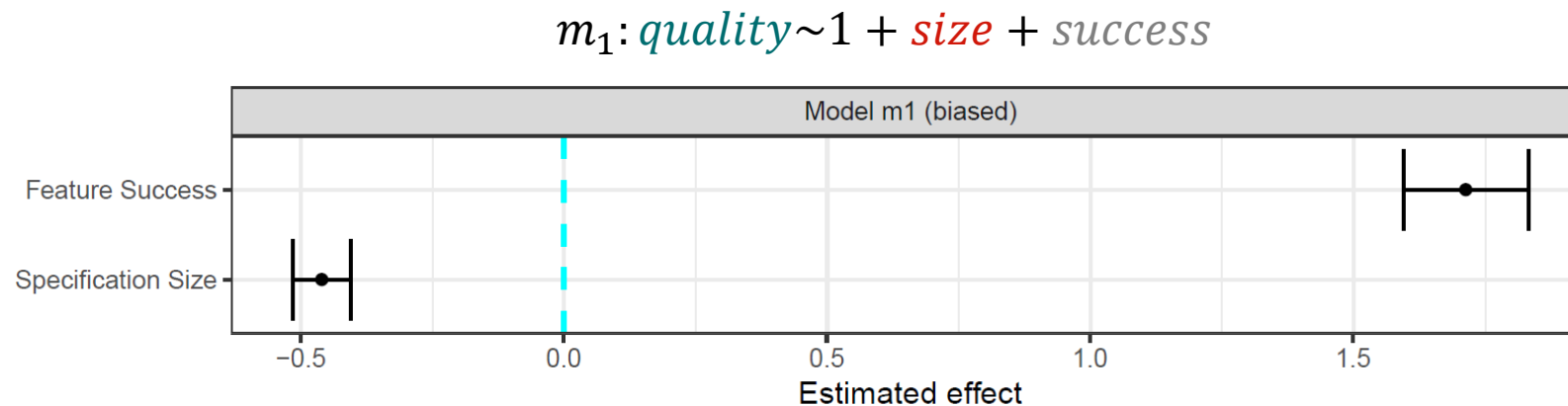
$m_2: \text{quality} \sim 1 + \text{size}$



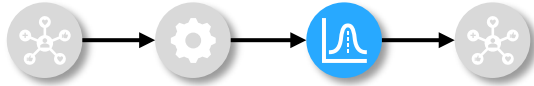
Colliding: Common Effects



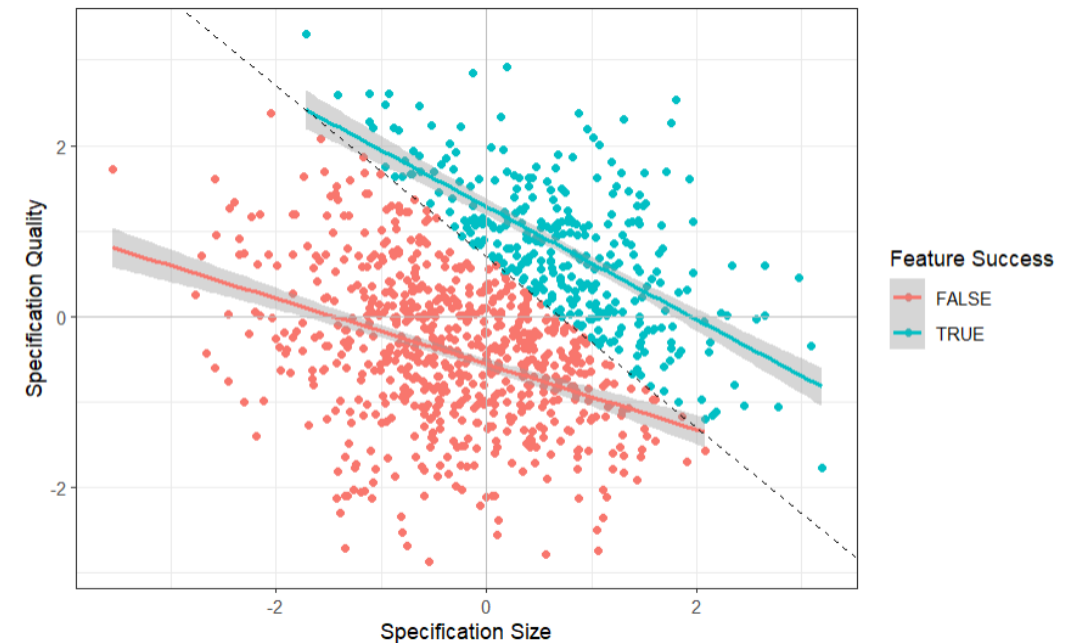
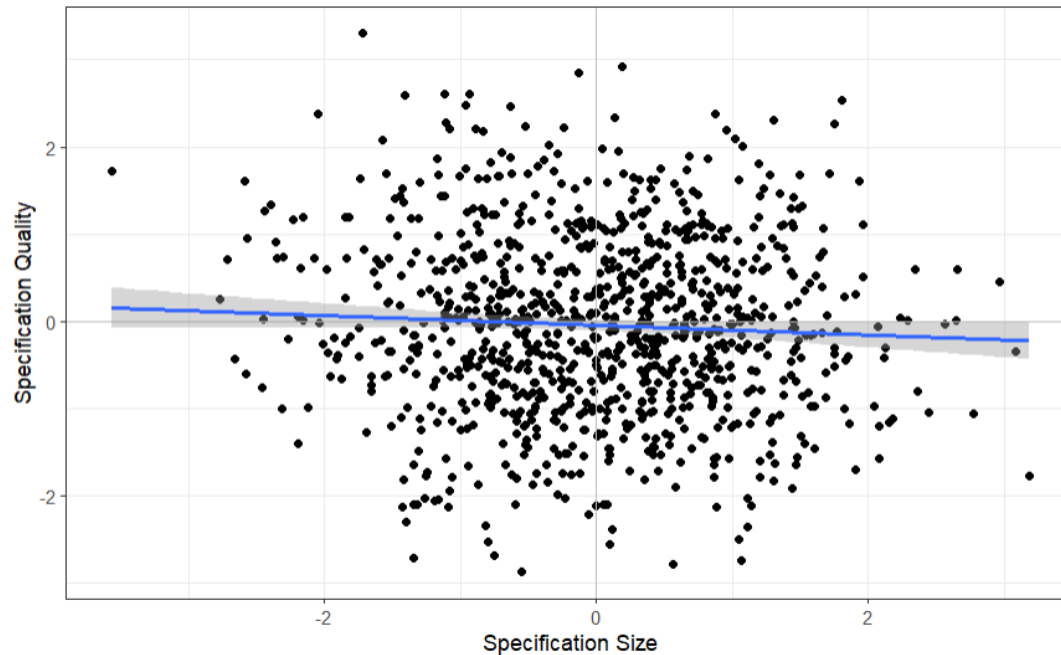
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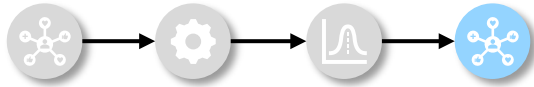
Colliding: Common Effects



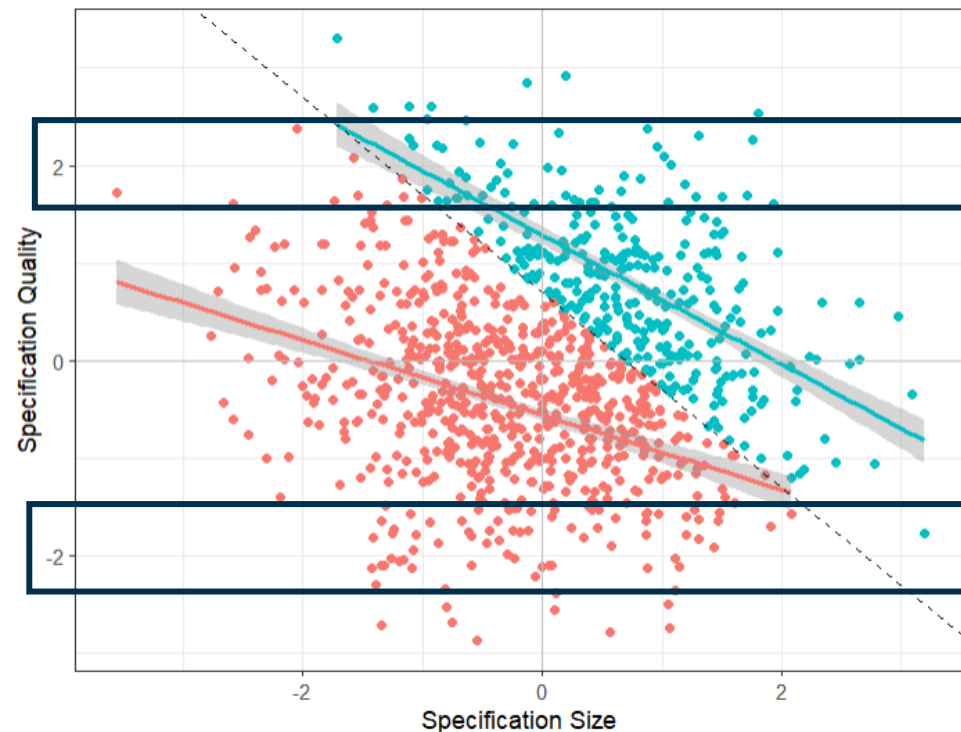
“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”



Colliding: Common Effects



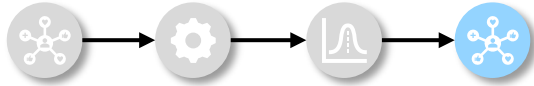
“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”



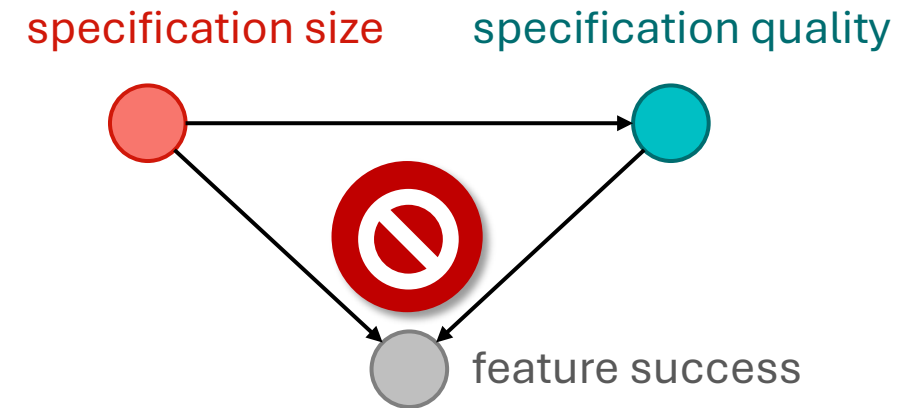
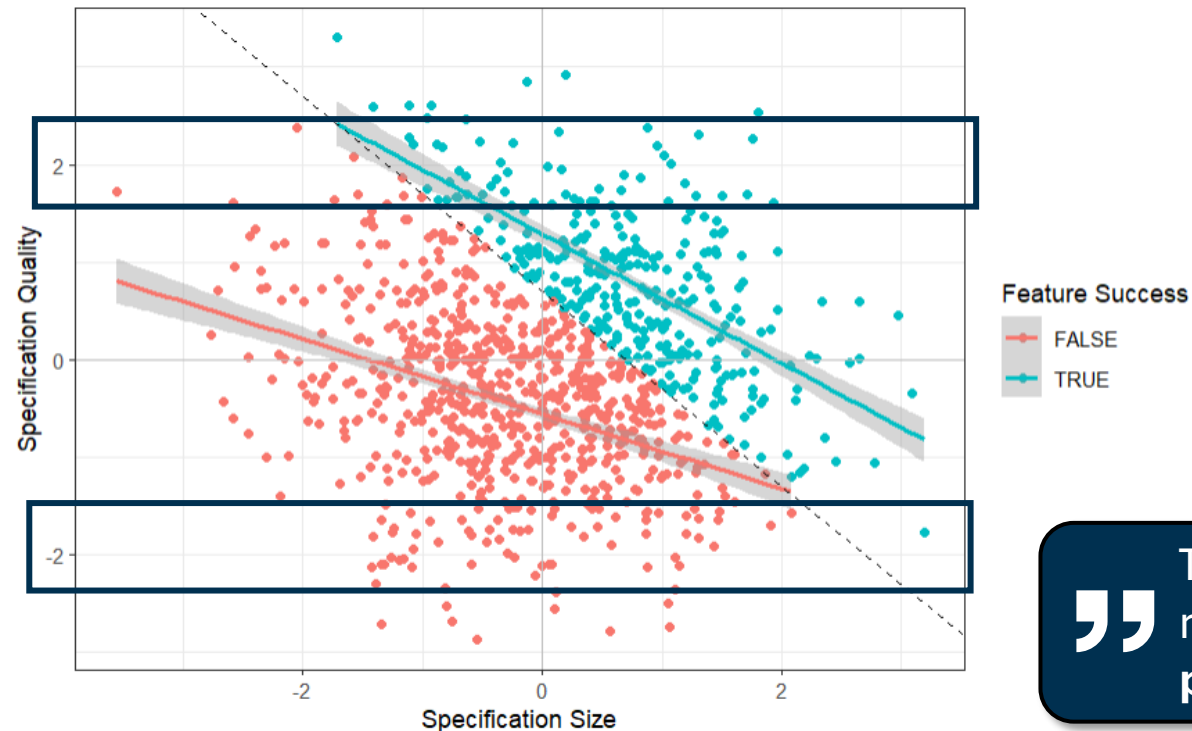
For any high **quality** value, if feature success is *false*, then it is more likely that the **specification size** was small.

For any low **quality** value, if feature success is *true*, then it is more likely that the **specification size** was large.

Colliding: Common Effects

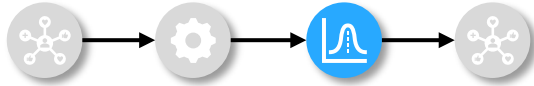


“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”

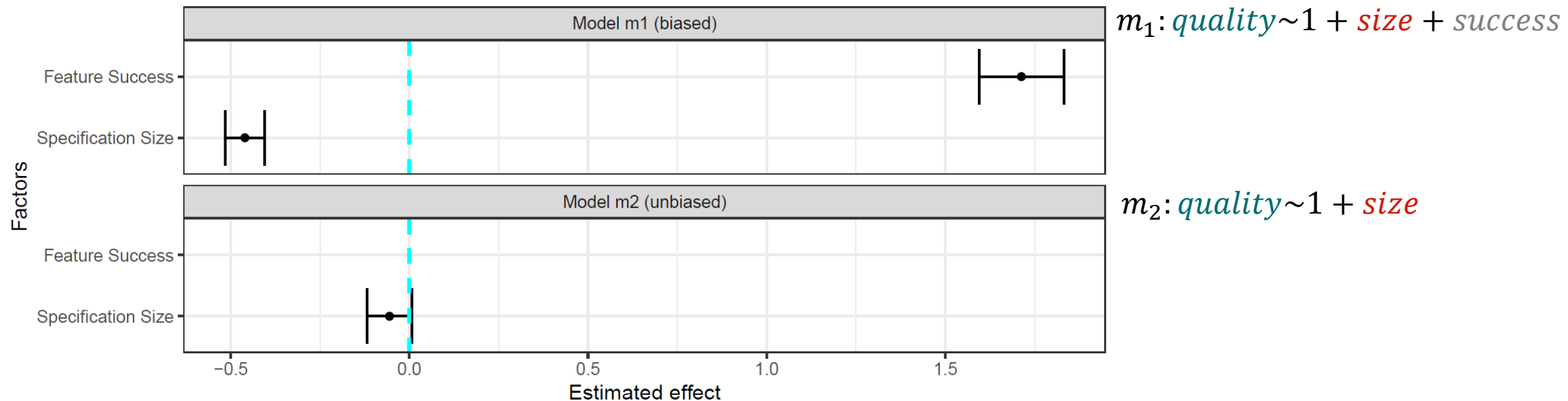


” The non-causal association suggests a negative relationship between the cause and effect (**Berkson’s paradox/fallacy**).

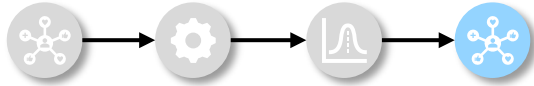
Colliding: Common Effects



“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”



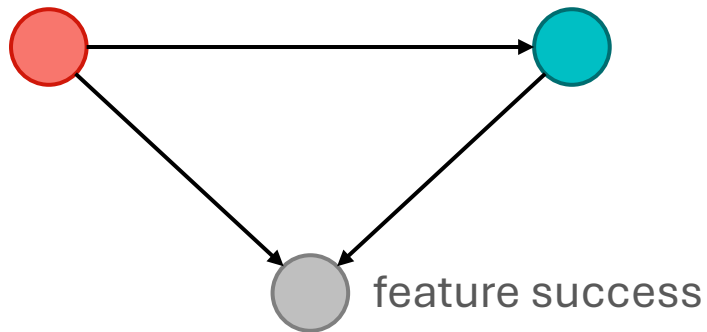
Colliding: Common Effects



“Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.”

specification size

specification quality



Model Derivation

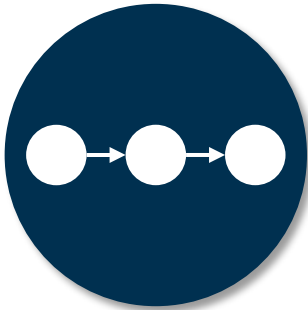
$$m_1: \text{quality} \sim 1 + \text{size} + \text{success}$$

$$m_2: \text{quality} \sim 1 + \text{size}$$

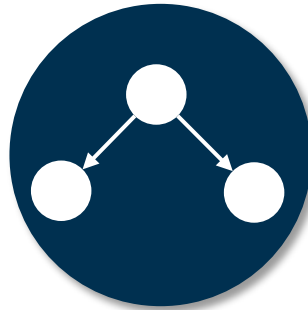
” If both **treatment** and **outcome** have a common effect, we must not condition on this common effect. Consequently, we cannot simply add *all* relevant variables as predictors.

Basic Types of Association

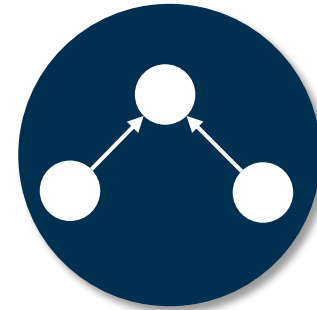
There are three basic types of association



Mediator



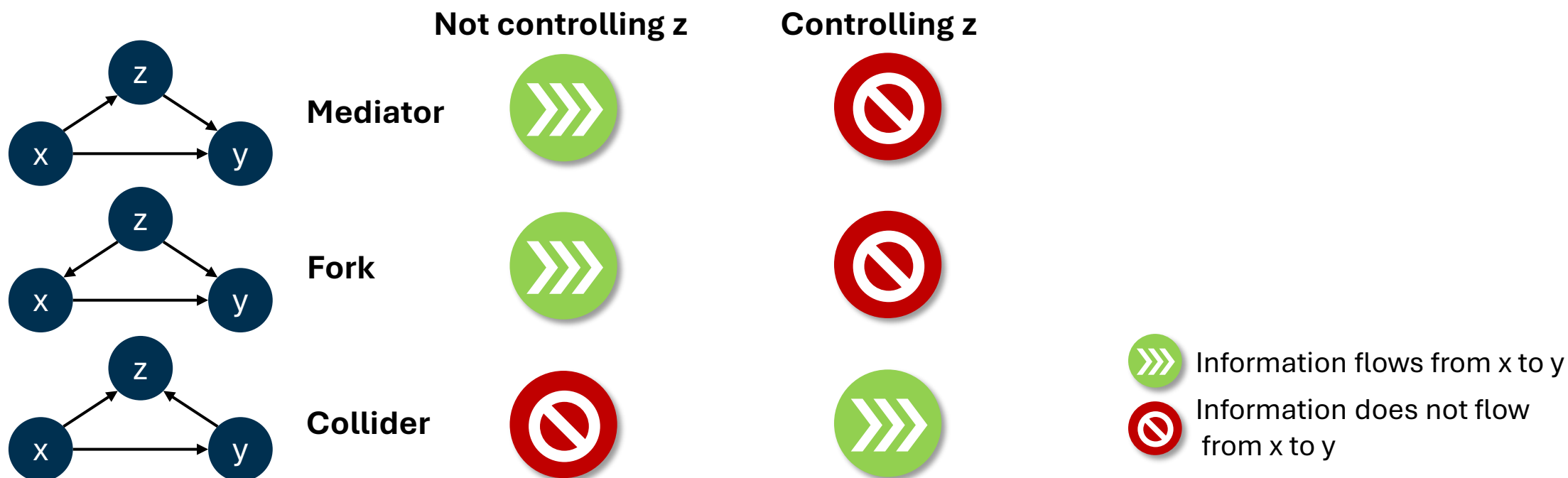
Fork



Collider

Basic Types of Association

The three basic types of association behave differently when statistically controlling for the third variable:

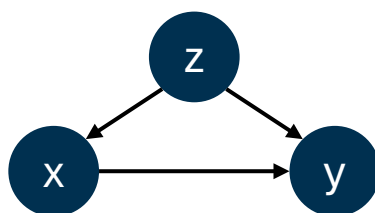


Causal Modelling

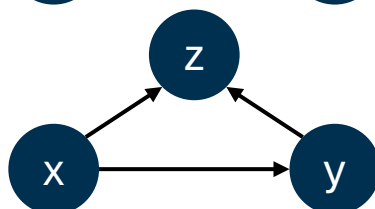
Causal modelling is imperative for drawing the correct conclusions when conducting statistical causal inference.

” We cannot infer causality from data alone – we must also know the process that created that data.

Causal Model



Fork



Collider



Model Derivation

x	y	z

Statistical Model

$$m: y \sim x + z$$

$$m: y \sim x$$

? But how does this approach scale beyond three variables?



d-Separation

Statistical Control for Reducing Bias

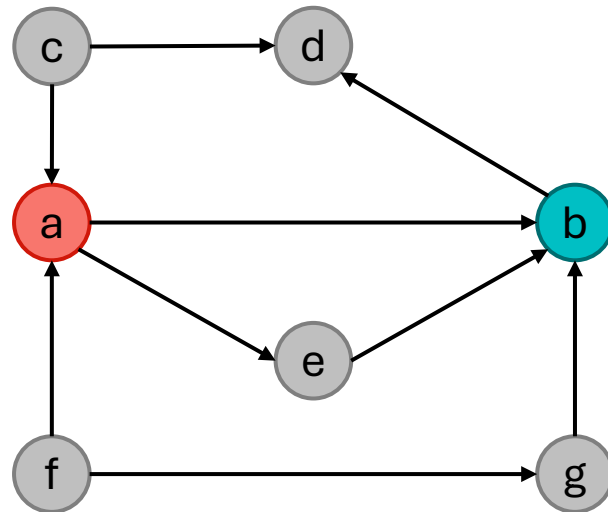
Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

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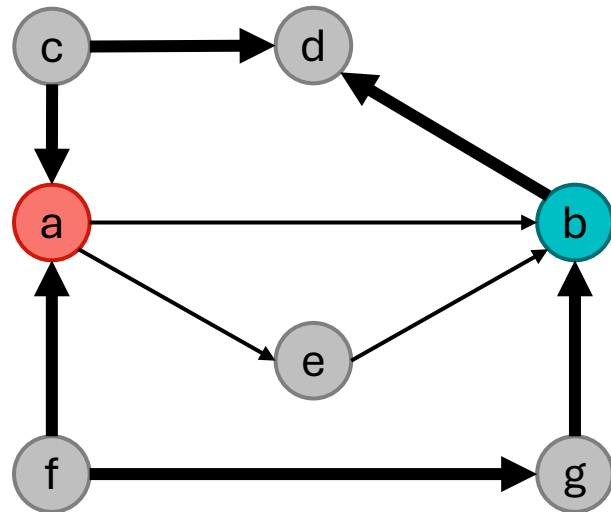
Model Derivation

$m: b \sim a$

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

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Model Derivation

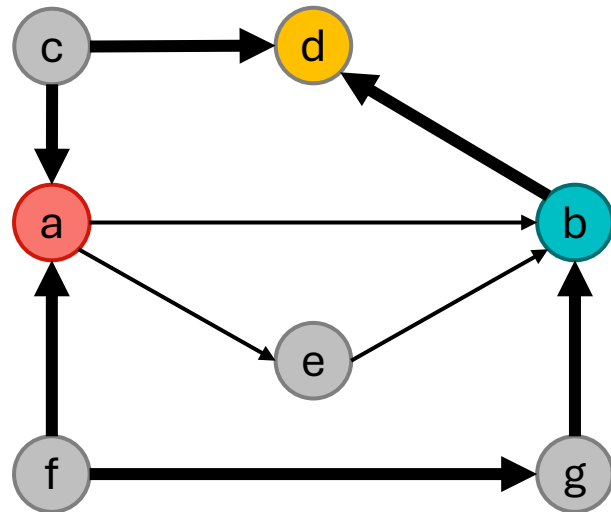
$m: b \sim a$

1. Identify every **backdoor path**.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

Deriving a statistical Model from a causal Model: d-separation

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Model Derivation

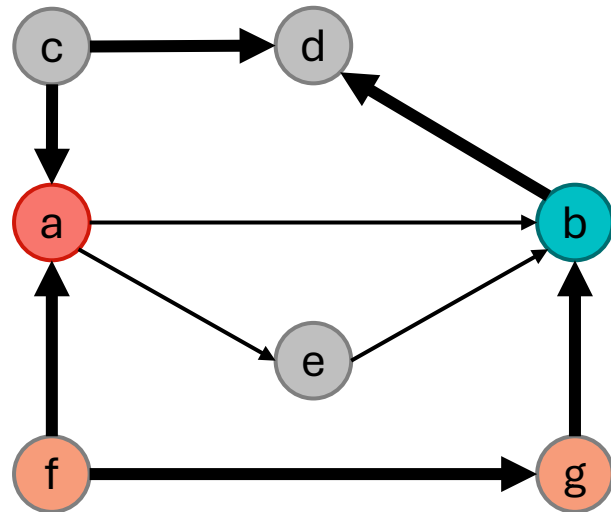
$m: b \sim a$

1. Identify every backdoor path.
2. Check if there is a **collider** between the cause and effect of interest.
 1. If yes, do not condition on any of the variables on this path.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



 Model Derivation

$$m: b \sim a + f$$

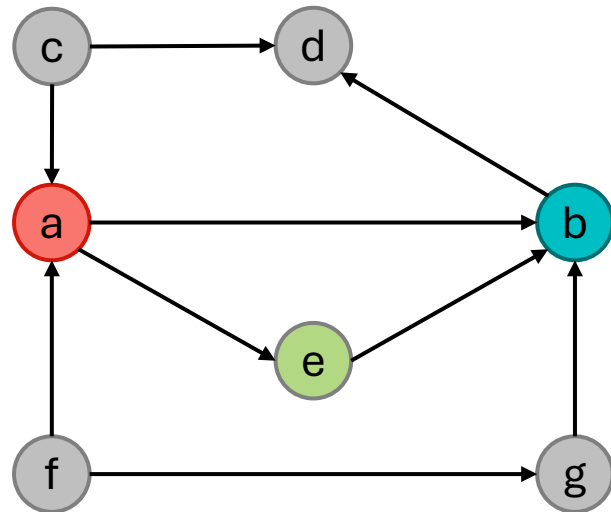
$$m: b \sim a + g$$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
 1. If yes, do not condition on any of the variables on this path.
 2. If not, condition on **any of the variables** on this path to “close” the backdoor path.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



Model Derivation

$$m: b \sim a + f(+e)$$

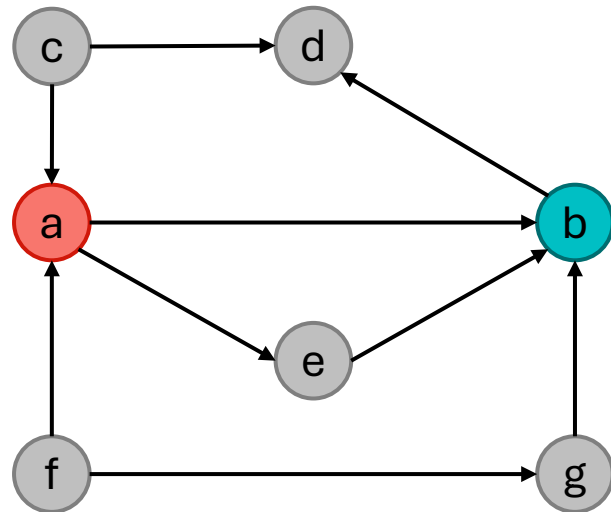
$$m: b \sim a + g(+e)$$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
 1. If yes, do not condition on any of the variables on this path.
 2. If not, condition on any of the variables on this path to “close” the backdoor path.
3. Optionally, condition on **mediators** to discern the direct effect.

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Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



Model Derivation

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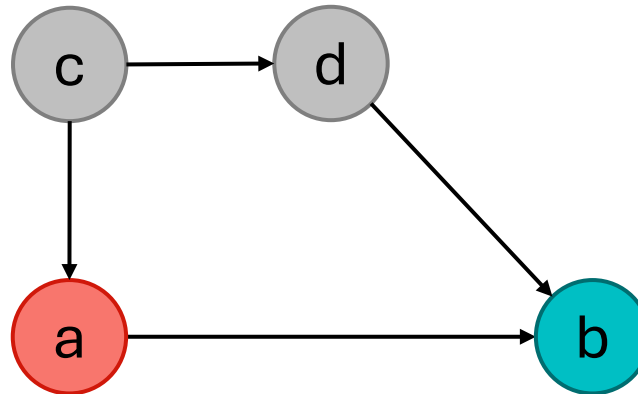
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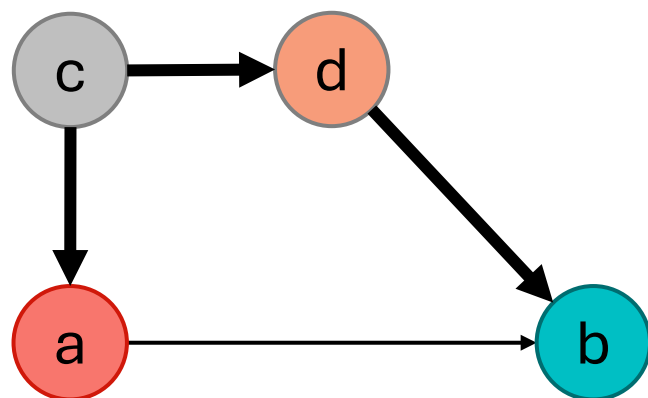
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.

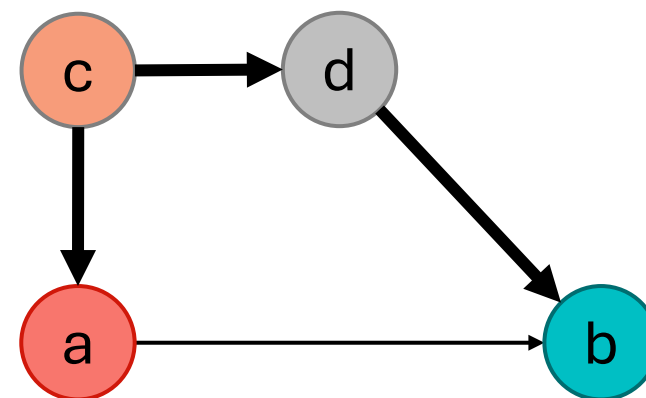


Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



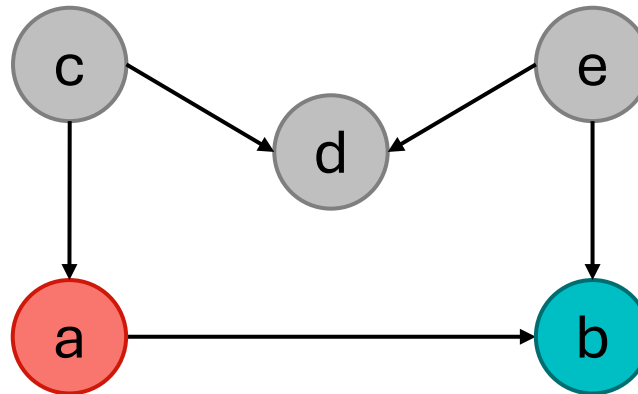
Adjustment set = {d}



Adjustment set = {c}

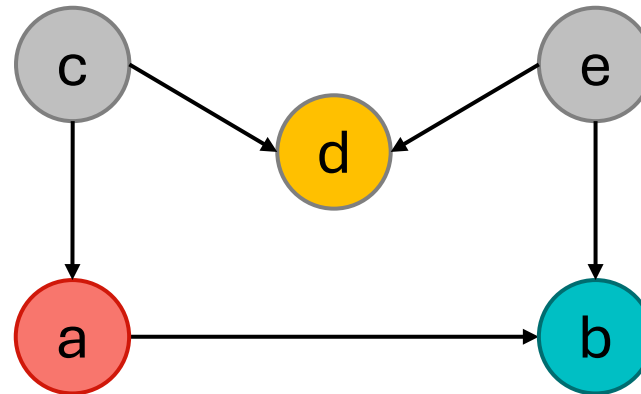
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Exercises

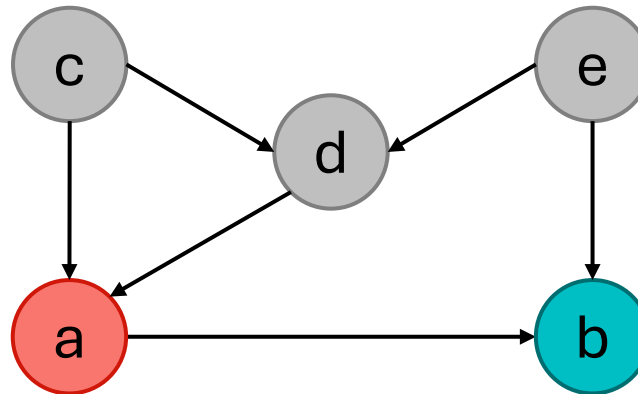
Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Adjustment set = $\{\}$

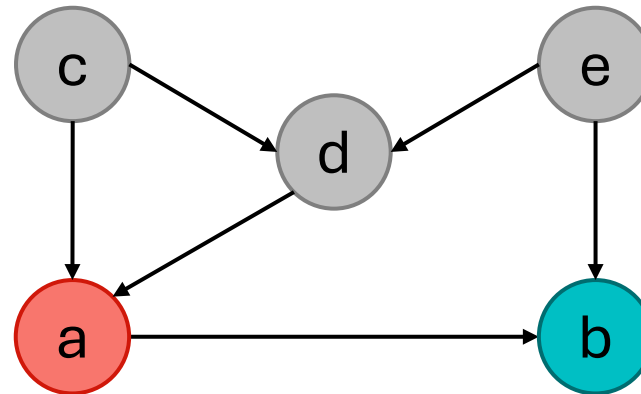
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.

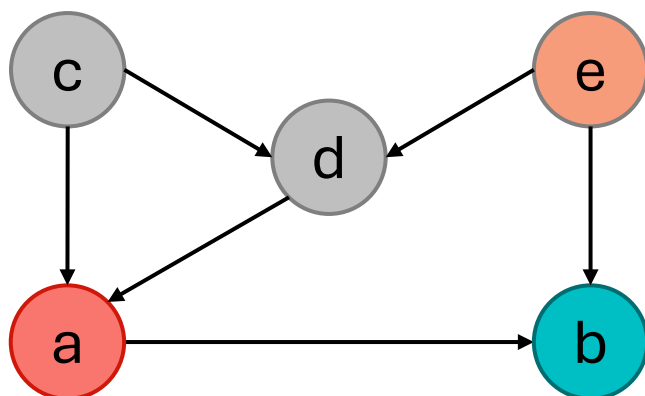


The DAG contains two backdoor paths:

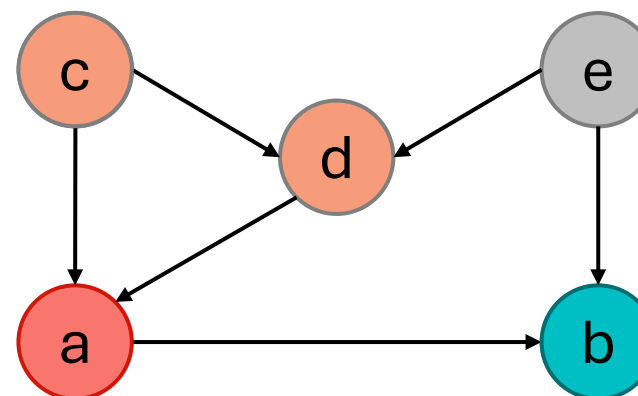
1. $a \leftarrow c \rightarrow d \leftarrow e \rightarrow b$
2. $a \leftarrow d \leftarrow e \rightarrow b$

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Adjustment set = {e}



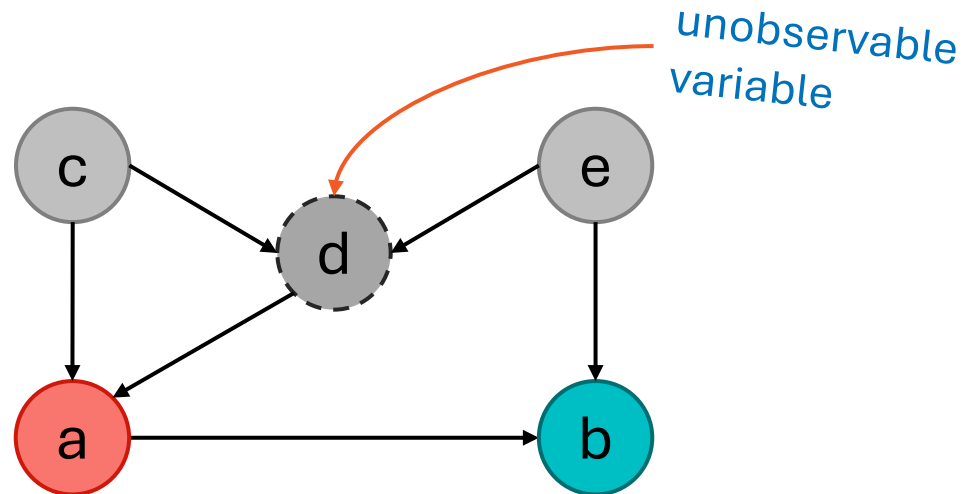
Adjustment set = {c, d}



Conditioning on d opens the backdoor-path $a \leftarrow c \rightarrow d \leftarrow e \rightarrow b$ which then has to be closed by additionally conditioning on c .

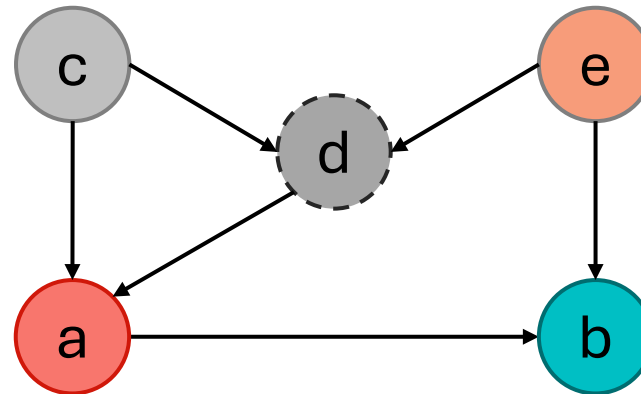
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Exercises

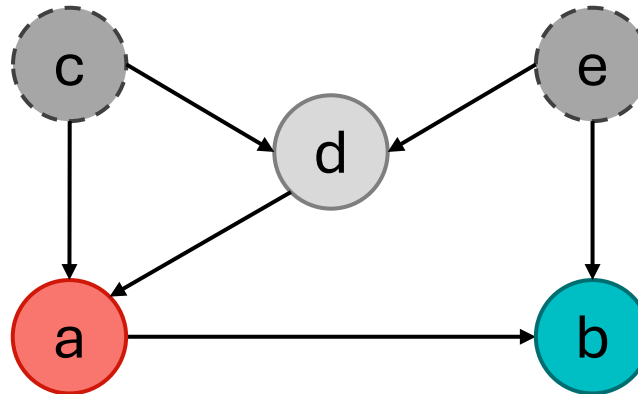
Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Adjustment set = {e}

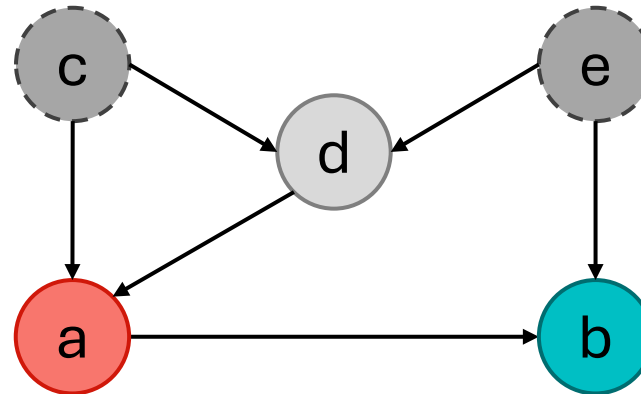
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



” For some DAGs, it is impossible to find an adjustment set that deconfounds the effect of interest.



Causal Workflow

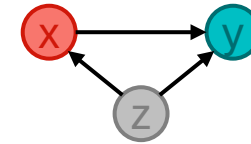
Applying Statistical Causal Inference in Research and Practice

Workflow for Statistical Causal Inference



1) Modelling

Creating a DAG of assumed causal relationships

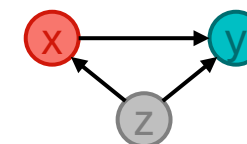


Workflow for Statistical Causal Inference



1) Modelling

Creating a DAG of assumed causal relationships

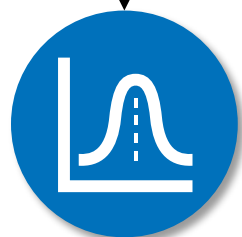


2) Identification

Deriving a statistical from the causal model

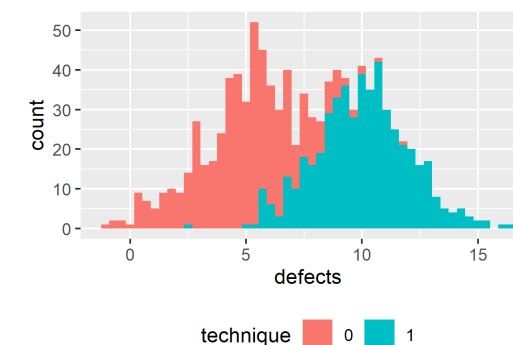
 Model Derivation

$$y \sim x + z$$



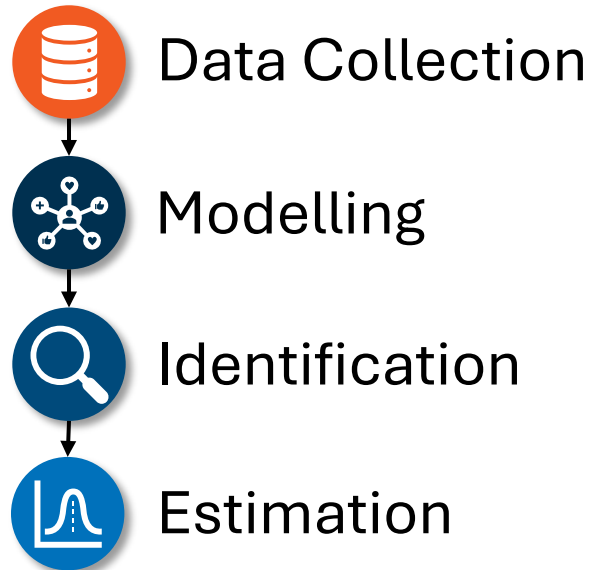
3) Estimation

Training a statistical model with the observed data and evaluating the results

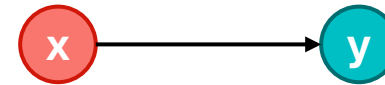


Siebert, J. (2023). Applications of statistical causal inference in software engineering. *Information and Software Technology*, 159, 107198.

Scientific Workflow: Traditional



Assumed DAG

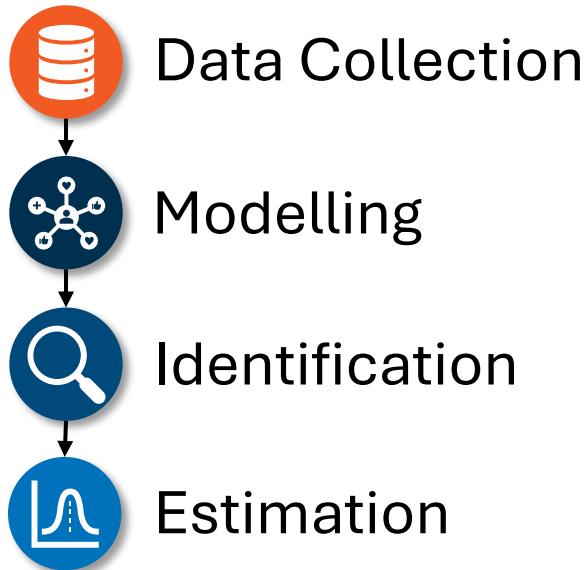


Collected Data

x	y

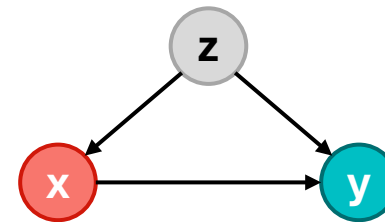
Too often, we derive hypotheses from the data available to us.

Scientific Workflow: Traditional



Assumed DAG

hard-to-measure
confounder

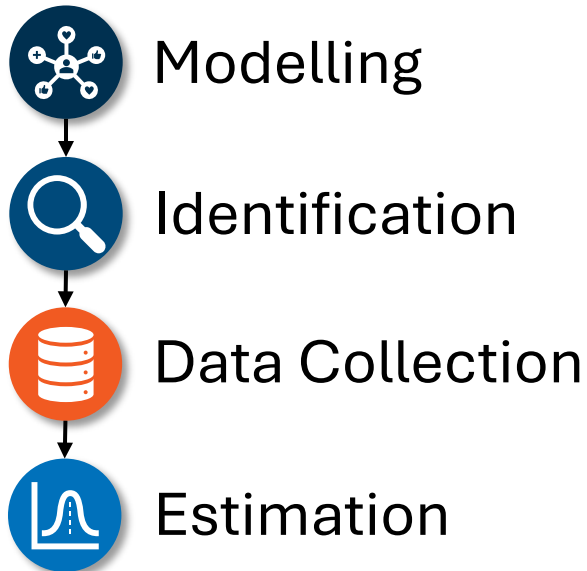


Collected Data

x	y

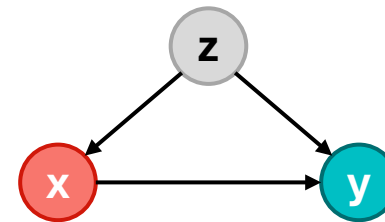
” Not collecting data about a confounder will threaten the internal validity of your result

Scientific Workflow: Better



Assumed DAG

hard-to-measure
confounder



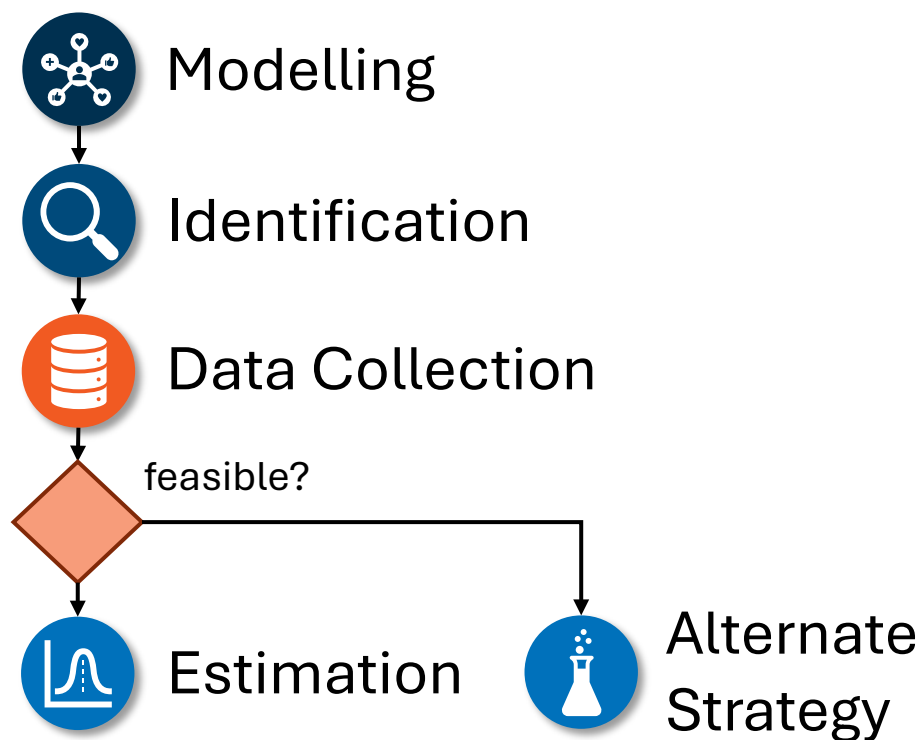
Collected Data

x	y	z



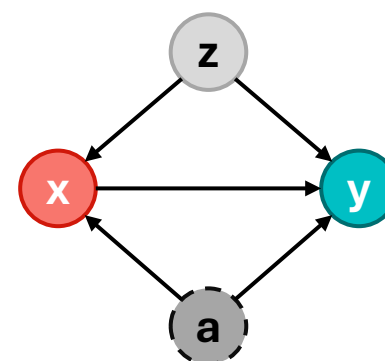
Determine the variables about which you need to collect data in order to estimate an unbiased effect of interest *before* collecting that data.

Scientific Workflow: Best



Assumed DAG

hard-to-measure
confounder



unobservable
confounder

Collected Data

x	y	z	a
			x
			x
			x
			x

Unobservable confounders require (a) acknowledgement in the **threats to (internal) validity** and (b) **different research strategies**.

Recommendations



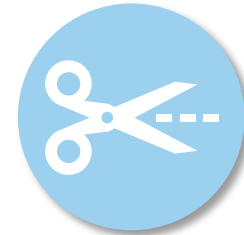
Data Collection

Is a part of data analysis,
not its precursor



Draw a DAG

representing your causal
assumptions



Use d-separation

to figure out (1) if you can
conduct your study and
(2) what data you need to
collect



Conclusion

Final Remarks

Reading list



Pearl, J., & Mackenzie, D. (2018). *The book of why: the new science of cause and effect*. Basic books.



McElreath, R. (2018). *Statistical rethinking: A Bayesian course with examples in R and Stan*. Chapman and Hall/CRC.



Cinelli, C., Forney, A., & Pearl, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, 53(3), 1071-1104.



Siebert, J. (2023). Applications of statistical causal inference in software engineering. *Information and Software Technology*, 159, 107198.



Furia, C. A., Feldt, R., & Torkar, R. (2019). Bayesian data analysis in empirical software engineering research. *IEEE Transactions on Software Engineering*, 47(9), 1786-1810.



Furia, C. A., Torkar, R., & Feldt, R. (2022). Applying Bayesian analysis guidelines to empirical software engineering data: The case of programming languages and code quality. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 31(3), 1-38.

Material

All material is public and open-source on GitHub. Feel free to clone, fork, and share the study material.

Comments, requests, and additions are welcome.

20.01.2026

JulianFrattini/**bda4sci** Public

The screenshot shows the GitHub repository page for JulianFrattini/bda4sci. The repository is public and has 4 stars, 1 watch, and 0 forks. The main branch is 'main'. The repository contains a README, a LICENSE, and a .gitignore file. The README is titled 'Introduction to Statistical Causal Inference' and describes the purpose of the seminar. The repository is licensed under Apache-2.0.

Repository Details:

- Owner: JulianFrattini
- Repository Name: bda4sci
- Visibility: Public
- Stars: 4
- Watchers: 1
- Forks: 0
- Unpin: Unpin
- Unwatch: Unwatch
- Star: Star

Files and Commits:

File	Commit Message	Commit Date
version1	Reorganized repository structure	5 months ago
.gitignore	Reorganized repository structure	5 months ago
LICENSE	Initial commit	11 months ago
README.md	Reorganized repository structure	5 months ago
sci-intro.Rproj	Reorganized repository structure	5 months ago

Repository Content:

Introduction to Statistical Causal Inference

license Apache-2.0

This repository contains the material for the introduction seminar to Statistical Causal Inference (SCI). The purpose of the seminar is to introduce software engineering researchers with a background in analysis of quantitative data to a causal framework for inferential statistics proposed by Judea Pearl^[1] and Richard McElreath^[2]

About:

Material for the Introduction Seminar to Bayesian Data Analysis for Statistical Causal Inference

bayesian-data-analysis
statistical-causal-inference

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Limitations

This was but a gentle introduction. The topic of statistical causal inference contains many more elements, like

- **Ancestors:** parent-child relationships in DAGs
- **Interaction:** moderated inter-relationship between two or more variables and an outcome
- **Hierarchical model:** partial pooling
- **Different types of regression:** regression models for different outcome variable distributions
- **Bayesian data analysis:** advanced estimation technique

And many more ...